Copernicus Global Land Operations
"Vegetation and Energy"
"CGLOPS-1"
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PRODUCT QUALITY ASSURANCE DOCUMENT

Issue I2.40

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10.09.2026	All	Overall update and addition of information related to the validation of Evapotranspiration products.	I2.40

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List of Acronyms

AATSR	Advanced Along-Track Scanning Radiometer
AC	Atmospheric Corrections
ACIX	Atmospheric Correction Intercomparison eXercise
AD	Applicable Document
AOD	Aerosol Optical Depth
AOI	Area Of Interest
AOT	Aerosol Optical Thickness
APAR	Absorbed Photosynthetically Active Radiation
APU	Accuracy Precision Uncertainty
ARM	Atmospheric Radiation Measurement
ASCAT	Advanced Scatterometer
ASCII	American Standard Code for Information Interchange
AVHRR	Advanced Very High-Resolution Radiometer
BELMANIP	Benchmark Land Multisite Analysis and Intercomparison of Products
BfG	German Federal Institute of Hydrology
BRDF	Bidirectional Reflectance Distribution Function
BRF	Bidirectional Reflectance Factor
C3S	Copernicus Climate Change Service
CAMS	Copernicus Atmosphere Monitoring Service
CCI	Climate Change Initiative
CCRS	Canada Center for Remote Sensing
CEOS	Committee on Earth Observation Satellites
CGLOPS	Copernicus Global Land OperationS
CLMS	Copernicus Land Monitoring Service
CSAR	C-band Synthetic Aperture Radar
CUL	Cultivated
DBF	Deciduous Broadleaf Forest
DM	Dry Matter
DMP	Dry Matter Productivity
EBF	Evergreen Broadleaf Forest
ECMWF	European Center for Medium range Weather Forecast
EEA	European Environmental Agency

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EO	Earth Observation
EOS	Earth Observation System
EOSD	End Of Season Day
EPS	EUMETSAT Polar System
ERA	ECMWF Atmospheric Reanalysis of the global climate
ESA	European Space Agency
ESU	Elementary Sampling Unit
ETA	Actual Evapotranspiration
ETC	Ecosystem Thematic Centre
EUMETSAT	European organisation for the exploitation of Meteorological Satellites
EVI	Enhanced Vegetation Index
FAPAR	Fraction of photosynthetically Active Radiation absorbed by the vegetation
FCover	Fraction of vegetation cover
FLUXNET	Network of Flux tower sites
FRM	Fiducial Reference Measurement
FRM4Veg	Fiducial Reference Measurement for Vegetation
FTP	File Transfer Protocol
FWC-2	Framework Contract n°2
GAUL	Global Administrative Unit Layers
GBA	Global Burned Area
GBOV	Ground-Based Observations for Validation
GCC	Green Chromatic Coordinates
GDMP	Gross Dry Matter Productivity
GIO	GMES Initial Operations
GL	Global Land
GLC	Global Land Cover
GLDAS	Global Land Data Assimilation System
GMR	Geometric Mean Regression
GOES	Geostationary Operational Environmental Satellite
GPP	Gross Primary Productivity
GSW	Generalized Split-Windows
GUI	Graphical User Interface
GUM	Guides to the expression of Uncertainty in Measurement
H	Sensible Heat Flux

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HER	Herbaceous
HF	Heat Fluxes
HR-VPP	High Resolution Vegetation Phenology and Productivity
ICOS	European Integrated Carbon Observation System
ICWRGC	International Center for Water Resources and Global Change
IGBP	International Geosphere-Biosphere Programme
IMS	Snow and Ice Mapping System
IODC	Indian Ocean Data Coverage
IPMA	Instituto Português do Mar e da Atmosfera
ISMN	International Soil Moisture Network
ISO	International Organization for Standardization
JCGM	Joint Committee for Guides in Metrology
JPSS	Joint Polar Satellite System
JRC	Joint Research Center
KIT	Karlsruhe Institute of Technology
LAI	Leaf Area Index
LANDVAL	Land Validation
LE	Latent Heat Flux
LER	Lambert Equivalent Reflectance
LPV	Land Product Validation group
LSA SAF	Land Surface Analysis of Satellite Applications Facility
LSM	Land Surface Model
LSP	Land Surface Phenology
LST	Land Surface Temperature
LST10-DC	Land Surface Temperature 10-daily Daily Cycle
LST10-SYN	Land Surface Temperature 10-daily Synthesis
LST10-TCI	Land Surface Temperature 10-daily Thermal Condition Index
LTER	Long Term Ecological Research network
LTS	Long Term Statistics
LUE	Light Use Efficiency
LUT	Look-Up-Tables
MAD	Median Absolute Deviation
MAR	Major Axis Regression
MARSOP	Monitoring Agriculture by Remote Sensing Operations

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MD	Median Deviation
MERIS	MEdium Resolution Imaging Spectrometer
METOP	Meteorological Operational Satellite
MGVI	MERIS Global Vegetation Index
MODIS	Moderate Resolution Imaging Spectroradiometer
MSG	METEOSAT Second Generation
MSI	Multi-Spectral Imager
MTSAT	Multi-Functional Transport Satellites
NASA	National Aeronautics and Space Agency
NBAR	Nadir BRDF-Adjusted Reflectance
NCAR	National Center for Atmospheric Research
NCEP	National Centers for Environmental Prediction
NDVI	Normalized Difference Vegetation Index
NEE	Net Ecosystem Exchange
NEON	National Ecological Observatory Network
netCDF	Network Common Data Form
NIR	Near Infra-Red
NLF	Needleleaf forest
NOBS	Number of Observations
NPN	National Phenology Network
NPP	Net Primary Productivity
NRT	Near real-Time
NTSG	Numerical Terradynamic Simulation Group
NWC-SAF	Satellite Application Facility on Support to Nowcasting and Very Short-Range Forecasting
OF	Other Forests
OGVI	OLCI Global Vegetation Index
OLCI	Ocean and Land Color Imager
OLIVE	Online Validation Exercise
OZFlux	Australian and New Zealand Flux Research Monitoring
PDF	Probability Density Function
PEM	Production Efficiency Model
PEP	Pan European Phenology Project
PGQ	Proportion of Good Quality
PPI	Plant Phenology Index

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QA	Quality Assessment
QA4EO	Quality Assurance for Earth Observations
QAR	Quality Assessment Report
QFLAG	Quality Flag
QM	Quality Monitoring
RadCalNet	Radiation Calibration Network
REBELS	Regularized Brdf inversion for Land Surface reflectances
Reco	Ecosystem Respiration
RM	Reference Measurements
RMSD	Root Mean Square Deviation
RMSE	Root Mean Square Error
ROI	Region Of Interest
RTM	Radiative Transfer Model
S10	10-day synthesis
S3_SY_2_SYN	ESA Sentinel-3 Level-2 SYNnergy product
SAFARI	Southern African Regional Science Initiative
SAR	Synthetic Aperture Radar
SAVS	Surface Albedo Validation Sites
SBA	Sparse vegetated and Bare Areas
SCATSAR	Scatterometer-Synthetic Aperture Radar
SCE	Snow Cover Extent
SDR	Surface Directional Reflectance
SEVIRI	Spinning Enhanced Visible and Infrared Imager
SHR	Shrubland
SLSTR	Sea and Land Surface Temperature Radiometer
SM	Soil Moisture
SMAC	Simplified Model for Atmospheric Correction
SNPP	Suomi National Polar-orbiting Partnership
SOSD	Start Of Season Day
SPOT	Satellite pour l'Observation de la Terre
SQE	Scientific Quality Evaluation
SRF	Spectral Response Functions
SSF	Surface State Flag
SM	Soil Moisture

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SSM	Surface Soil Moisture
ST	Seasonal Trajectories
STD	Standard Deviation
SWI	Soil water Index
SWIR	ShortWave Infra-Red
TCI	Thermal Condition Index
TIMESAT	a program for analyzing time-series of satellite sensor data
TIP	Two-stream Inversion Package
TOA	Top Of Atmosphere
TOC	Top Of Canopy
TOC-r	normalized Top Of Canopy Reflectance
TUWien	Technical University of Vienna
US	United States
USA	United States of America
VGT	SPOT/VEGETATION
VALERI	Validation of Land European Remote Sensing Instrument
VI	Vegetation Index
VIIRS	Visible/Infrared Imager Radiometer Suite
VITO	Flemish institute for technological research
VNIR	Visible and Near InfraRed
VPP	Vegetation Phenology and Productivity
VR	Validation Report
WGCV	Working Group on Calibration and Validation

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GENERAL DEFINITIONS

Accuracy is the degree of the "closeness of the agreement between the result of a measurement and a true valued of the measurand" [JCGM-GUM]. Commonly, accuracy represents systematic errors and often is computed as the statistical mean bias, i. e. the difference between the short-term average measured value of a variable and the true value. The short-term average is the average of a sufficient number of successive measurements of the variable under identical conditions, such that the random error is negligible relative to the systematic error. The latter can be introduced by instrument biases or through the choice of remote sensing retrieval schemes [GCOS-143].

Dry Matter Productivity (DMP) is an indicator designed to represent the growth rate of dry biomass: it is the amount (weight) of dry matter produced per surface unit and per time unit. The accumulation over time leads to the assessment of the biomass (in this context, phytomass), herein defined as above-ground mass per unit area of living plant material [GCOS-244]. The DMP is equivalent to the **Net Primary Productivity** (NPP) but has been adapted to agro-statistical use by transformation of the units into kilograms of dry matter produced per hectare and per day (kg DM ha^{-1} day^{-1}). Similar to the relation between NPP and DMP also the agronomic equivalent of the **Gross Primary Productivity** (GPP) can be defined as **Gross Dry Matter Productivity** (GDMP). The main difference between the NPP/DMP and GPP/GDMP is the inclusion of the autotrophic respiration.

Essential Climate Variable (ECV) is a variable or group of related variables that critically contribute to the characterization of Earth's climate state and forcing.

Evapotranspiration (ET) is the process by which water is transferred from the land to the atmosphere by evaporation from the soil and other surfaces and by transpiration from plants**.**

Fraction of Absorbed Photosynthetically Active Radiation (FAPAR) is defined as the fraction of Photosynthetically Active Radiation (PAR; solar radiation reaching the surface in the 400-700 nm spectral region) that is absorbed by a vegetation canopy [GCOS-200].

Fraction of green Vegetation Cover (FCover) is defined as the fraction of ground surface covered by green vegetation as seen from the nadir direction (Baret et al. 2013).

Land Surface Phenology (LSP) describes vegetation changes observed from satellite remote sensing and constitutes therefore an aggregate signal from heterogeneous surface conditions that may not be representative of any single plant species response (Kimball 2014).

Land Surface Temperature (LST) is a measure of how hot or cold the uppermost surface of the Earth is. For ground-based, airborne and space-borne remote sensing instruments, it is the aggregated radiometric temperature of the ensemble of components within the sensor field of view [GCOS-200].

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Latent heat flux is the rate of energy transfer between the Earth surface and the atmosphere due to water changing phase in the process of evapotranspiration or condensation.

Leaf Area Index (LAI) is defined as the total one-sided area of all leaves in the canopy within a defined region, and is a non-dimensional quantity, although units of [m^2/m^2] are often quoted, as a reminder of its meaning [GCOS-200].

Net Primary Production is the rate at which plants produce net useful chemical energy expressed in mass of carbon per unit area per day ($gC/m^2/day$).

Normalized Difference Vegetation Index (NDVI) is the difference between maximum (in NIR) and minimum (round the Red) vegetation reflectance, normalized to the summation. The NDVI is an indicator of the greenness of vegetation.

Precision or repeatability is the "closeness of agreement between the results of successive measurements of the same measurand carried out under the same conditions of measurement" [JCGM-GUM].

Sensible heat flux is the rate of transfer of heat between the Earth surface and the atmosphere which results in temperature change of the atmosphere.

Stability is the change in bias (apparent error) over a time period (Merchant, 2013) [GCOS-245].

Surface Soil Moisture is defined as the relative water content of the top few centimetres soil, describing how wet or dry the soil is in its topmost layer, expressed in percent saturation. It is measured by satellite radar sensors and allows insights in local precipitation impacts and soil conditions.

Soil Water Index quantifies the moisture condition at various depths in the soil. It is mainly driven by the precipitation via the process of infiltration. Soil moisture is a very heterogeneous variable and varies on small scales with soil properties and drainage patterns. Satellite measurements integrate over relative large-scale areas, with the presence of vegetation adding complexity to the interpretation.

Uncertainty is a "parameter, associated with the result of a measurement that characterizes the dispersion of the values that could reasonably be attributed to the measurand" [JCGM-GUM]. Uncertainty includes systematic and random errors.

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EXECUTIVE SUMMARY

The Copernicus Land Monitoring Service (CLMS) produces a series of qualified bio-geophysical products on the status and evolution of land surface at global scale. The products are used to monitor vegetation, crops, water cycle, energy budget and terrestrial cryosphere. Production and delivery of the parameters take place in a timely manner and are complemented by the constitution of long-term time series.

Quality Assurance constitutes the mean of guaranteeing the compliance of products with user requirements, and new products must pass an exhaustive scientific evaluation before to be delivered to the users. In the Copernicus Global Land Operations Service, the quality assurance includes two aspects: (1) the exhaustive validation of the bio-geophysical products following, as much as possible, the CEOS LPV guidelines for validation protocols (https://lpvs.gsfc.nasa.gov/); (2) the routine (operational) monitoring of the continued delivered products throughout the service. The methodologies applied for the exhaustive validation of the products are presented in the Product Quality Assurance Document (PQAD) while the operational quality monitoring is described in the System Quality Assurance Document (SQAD).

This PQAD details the validation approaches followed to assess the quality of the vegetation variables, like the Leaf Area Index (LAI), the fraction of photosynthetically active radiation absorbed by the vegetation (FAPAR), the fraction of green vegetation cover (FCover), the (gross) dry matter productivity (GDMP/DMP), the Gross and Net Primary Production (GPP, NPP), the burned areas, the Normalized Difference Vegetation Index (NDVI) and the Land Surface Phenology (LSP), the soil moisture variables like the Surface Soil Moisture (SSM) and the Soil Water Index (SWI), the energy variables like the Land Surface Temperature (LST), the actual evapotranspiration (ETA) and the heat fluxes (HF). The document describes the datasets, in-situ, satellites, and model outputs, used as references and the various criteria and metrics of evaluation.

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1 BACKGROUND OF THE DOCUMENT

The Product Quality Assurance Document focuses only on the scientific validation methods while the operational quality monitoring, performed routinely at the production centers, is described in the System Quality Assurance Document.

1.1 SCOPE AND OBJECTIVES

This PQAD presents the procedures applied by the consortium (CGLOPS-1) in charge of the production of the "Vegetation and Energy" variables to check the quality of these products before dissemination.

This document does not include the procedures applied for the quality check of the "Water and Cryosphere" variables, which are under the responsibility of the CGLOPS-2 consortium.

1.2 CONTENT OF THE DOCUMENT

Chapter 2 presents the global CLMS "Vegetation and Energy" products.

Chapter 3 introduces the overall CGLOPS quality assurance strategy.

Chapter 4 describes the reference datasets.

Chapter 5 details the validation procedures for each (group of) products.

1.3 RELATED DOCUMENTS

1.3.1 Applicable documents

AD1: Part 2: Technical specifications of Framework Service Contract – Operation of the bio-geophysical variables systematic monitoring of the Global Land Component of the Copernicus Land Service 'CGLOPS' JRC/2023/OP/0273, 19th April 2023.

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1.3.2 Input documents

Document Id	Description
CGLOPS-1_SQAD	System Quality Assurance Document of the Copernicus Global Land Operations Service "Vegetation & Energy".
ATBDs	
CGLOPS1_ATBD_S3-AC-V1.1	Algorithm Theoretical Basis Document of the Version 1.1 of the atmospheric corrections applied on Sentinel-3 OLCI and SLSTR data.
CGLOPS1_ATBD_LAI[FAPAR, FCOVER]300m-V1	Algorithm Theoretical Basis Document of 300m LAI, FAPAR, FCOVER Version 1 product, derived from PROBA-V data
CGLOPS1_ATBD_LAI[FAPAR, FCOVER]300m-V1.1	Algorithm Theoretical Basis Document of 300m LAI, FAPAR, FCOVER Version 1.1 product, derived from Sentinel-3/OLCI data
CGLOPS1_ATBD_LAI[FAPAR, FCOVER]300m-V2	Algorithm Theoretical Basis Document of 300m LAI, FAPAR, FCOVER Version 2 product.
GIOGL1_ATBD_DMP1km-V1	Algorithm Theoretical Basis Document of 1km Dry Matter Productivity Version 1 product.
CGLOPS1_ATBD_DMP1km-V2	Algorithm Theoretical Basis Document of 1km Dry Matter Productivity Version 2 product.
CGLOPS1_ATBD_DMP300m-V1	Algorithm Theoretical Basis Document of 300m Dry Matter Productivity Version 1 product, derived from PROBA-V
CGLOPS1_ATBD_DMP300m-V1.1	Algorithm Theoretical Basis Document of 300m Dry Matter Productivity Version 1.1 product, derived from Sentinel-3/OLCI data
CGLOPS1_ATBD_DMP300m-V2	Algorithm Theoretical Basis Document of 300m Dry Matter Productivity Version 2 product.
CGLOPS1_ATBD_NDVI300m-V3	Algorithm Theoretical Basis Document of 300m Normalized Difference Vegetation Index Version 3 product.
CGLOPS1_ATBD_BRDFCorrection300m-V1.6	Algorithm Theoretical Basis Document of the Version 1.6 of the correction of the Bidirectional

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	Reflectance Distribution Function for 300m products.
CGLOPS1_ATBD_LSP300m-V2	Algorithm Theoretical Basis Document of Land Surface Phenology 300m Version 2.0 product.
CGLOPS1_ATBD_BA300m-V4.0	Algorithm Theoretical Basis Document of 300m Burned Area Version 4.0 product.
CGLOPS1_ATBD_LST-V3.0	Algorithm Theoretical Basis Document of Land Surface Temperature Version 3.0 product.
CGLOPS1_ATBD_LST10-V3.0	Algorithm Theoretical Basis Document of 10-day Land Surface Temperature Version 3.0 product
CGLOPS1_ATBD_NDVI300m-V2	Algorithm Theoretical Basis Document of 300m Normalized Difference Vegetation Index Version 2 product derived from Sentinel-3/OLCI data
CGLOPS1_ATBD_NDVI1km-V3	Algorithm Theoretical Basis Document of 1km Normalized Difference Vegetation Index Version 3 product
Quality assessment and validation reports	
CGLOPS1_QAR_S3-AC-V1.1	Quality Assessment Report of the version 1.1 of atmospheric corrections applied on Sentinel-3 OLCI and SLSTR data.
CGLOPS1_VR_S3-TOCr300m-v2.3	Validation Report of the Sentinel-3 OLCI and SLSTR TOC reflectances Version 2.3.
CGLOPS1_VR_LAI[FAPAR,FCOVER]300m-V2.0	Validation Report of the 300m LAI, FAPAR, FCOVER Version 2.0 products.
CGLOPS1_VR_DMP-NPP300m_V2	Validation Report of the 300m DMP/GDMP/NPP/GDPP Version 2 product.
CGLOPS1_QAR_NDVI300m-V3	Quality Assessment Report of the 300m Normalized Difference Vegetation Index Version 3 product.
CGLOPS1_VR_LSP300m-V2	Validation Report of the 300m Land Surface Phenology Version 2 product.
CGLOPS1_VR_BA300m-V4.0	Validation Report of 300m Burned Area Version 4.0 products.

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CGLOPS1_VR_ETA300m-V1	Validation report Evapotranspiration and Heat Flux 300m Version 1.0
CGLOPS1_VR_LST-LST10-V3.0	Validation Report of Land Surface Temperature Version 3.0 products.
CGLOPS1_VR_SSM1km-V2	Validation Report of the Surface Soil Moisture 1km Version 2.0 product.
CGLOPS1_VR_SWI1km-V3	Validation Report of the Soil Water Index 1km Version 3.0 product.
CGLOPS1_VR_SWI-V4	Validation Report of the Soil Water Index 0.1° Version 4.0 product.
CGLOPS1_QAR_NDVI1km-V3	Quality Assessment Report of the 1km Normalized Difference Vegetation Index Version 3 product

Available at https://land.copernicus.eu/en/technical-library

1.3.3 External documents

Document Id	Description
JCGM-GUM	Joint Committee for Guides in Metrology – Guides to the expression of Uncertainty in Measurement. ISO/IEC Guide 98 - Part 3, 2008.
Available at https://www.iso.org/sites/JCGM/GUM-introduction.htm	
GCOS-143	Guideline for the Generation of Datasets and Products Meeting GCOS Requirements, WMO 2010.
GCOS-200	The Global Observing System for Climate: Implementation Needs, WMO 2016
GCOS-244	The 2022 GCOS Implementation Plan
GCOS-245	The 2022 GCOS ECVs Requirements
Available at https://library.wmo.int	
ATBD_IdePix	Sentinel-3 OLCI IdePix (Identification of Pixel features) – Algorithm Theoretical Basis Document, Version 1.0, 4th May 2022
Available at https://zenodo.org/records/6517333	

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ATBD-GBOV-LP3-LP4-LP5	Ground-Based Observations for Validation - Algorithm Theoretical Basis Document - Vegetation Products: LP3 (LAI), LP4 (FAPAR) and LP5 (FCOVER).
GBOV-ATBD-RM2-RM8-RM9_LST	Algorithm Theoretical Basis Document – Land Surface Temperature products RM2 (Thermal radiation), RM8 (Emissivity) and RM9 (Surface temperature)

Available at https://land.copernicus.eu/en/products/GBOV

NEON.DOC.014040	NEON TOS Protocol and procedure: Plant Phenology

Available at https://data.neonscience.org/documents/10179/2139401/NEON.DOC.014040vD/6a4f81a7-20b5-ba40-ad63-7541ffe959d9?version=1.0

Fire_cci_D4.1.1_PVR_v2.1	Product Validation Report - CCI Fire Disturbance Phase 2, Version 2.1

Available at https://climate.esa.int/media/documents/Fire_cci_D4.1.1_PVR_v2.1.pdf

S2-PDGS-MPC-ATBD-L2A	Sentinel-2 MSI—Level 2A Algorithm Theoretical Basis Document, S2-PDGS-MPC-ATBD-L2A, Issue 2.10, 15.11.2021

Available at https://sentiwiki.copernicus.eu/web/s2-documents

OMPC.ACR.HBK.003	Sentinel-3 SYN Land User Handbook 2024, Issue 1.30, 05.08.2024
S3.PN.SYN-L2.09	Sentinel-3 Product Notice – SYNergy. Issue/rev date 18-07-2023
ATBD-GIFAPAR	Sentinel-3 OLCI GI-FAPAR & rectified channels in O10, O11, O12 and O17 bands with uncertainties – Algorithm Theoretical Basis Document, 2024. doi:10.2760/5540982

Available at https://sentiwiki.copernicus.eu/web/s3-documents

S3-L2-AOD-SYN-ATBD	AOD-SYN Algorithm Theoretical Basis Document, Version 1.12, last modified 2019-04-04

Available at https://sentinels.copernicus.eu/documents/247904/3618136/S3-L2-AOD-SYN-ATBD-v112.pdf/36dd4812-10c1-2818-74a4-2637f948149d?t=1664559347721

S3MPC.UOL.VR.029	Sentinel-3 SLSTR Level 2 Land Validation Report, 30th June 2017

Available at https://sentiwiki.copernicus.eu/__attachments/a_d23752b734dedbc6074fdbe2f96a0e254b896d1ef2a32adcbc978eac0609b7b0/Sentinel-3-SLSTR-Level-2-Land-Validation-Report.pdf?cb=73f44dd533c54e7839175a8c7172a31a

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ATBD-EPSVEGA	Algorithm Theoretical Basis Document for EPS-AVHRR Vegetation parameters (ETFVC, ETLAI, ETFAPAR), Ref: SAF/LAND/UV/ATBD_EPSVEGA/1.0, 10/07/2017
VR-EPSVEGA	Validation Report EPS/AVHRR - Vegetation Parameters (VEGA) LSA-403 (ETFVC), LSA-406 (ETLAI), LSA-409 (ETFAPAR) SAF/LAND/UV/VR_EPSVEGA/1.2, Issue 1, 30 July 2019
VR-EPSVEGA-R	Validation Report EPS/AVHRR Vegetation Parameters Data Record (VEGA-R)

Available at https://lsa-saf.eumetsat.int/en/user-support/documentation/

ATBD-VIIRS-LAI_FPAR	Algorithm Theoretical Basis Document for VIIRS LAI and FPAR, VNP15 LAI/FPAR ATBD Version 1.1, 17th April 2018.

Available at https://viirsland.gsfc.nasa.gov/PDF/VIIRS_LAI_FPAR_ATBD_V1.1.pdf

ATBD-MOD15	MODIS leaf area index (LAI) and fraction of photosynthetically active radiation absorbed by vegetation (FPAR) product (MOD15) - Algorithm Theoretical Basis Document, Version 4.0, 30th April 1999, NASA Goddard Space Flight Center, Greenbelt, MD, 20771

Available at https://modis.gsfc.nasa.gov/data/atbd/atbd_mod15.pdf

ATBD-C3S_V3	Copernicus Climate Change Service. Algorithm Theoretical Basis Document – Multi-sensor CDR LAI and fAPAR v3.0. Ref: C3S_D312b_Lot5.1.4.4-v3.0_ATBD_CDR-ICDR_LAI_FAPAR_MULTI_SENSOR_v3.0_PRODUCTS_v1.0.1
PQAR-C3S_V3	Copernicus Climate Change Service. Product Quality Assessment Report – Multi-sensor CDR LAI and fAPAR v3.0. Ref: D2.3.9-v3.0_PQAR_CDR_LAI_fAPAR_PROBAV_v3.0_PRODUCTS_v1.1
ATBD-C3S_V4.1	Copernicus Climate Change Service. Algorithm Theoretical Basis Document – CDR and ICDR Sentinel-3 LAI and FAPAR 300m v4.1. Ref: WP2-FDDP-LAI-FAPAR-SENTINEL3-300m-v4.1_ATBD_v1.0
PQAR-C3S_V4.1	Copernicus Climate Change Service. Product Quality Assessment Report – CDR/ICDR Sentinel-3 LAI and fAPAR v4.1. Ref: WP2-FDDP-LAI-FAPAR-SENTINEL3-300m-v4.1_PQAR_v1.1

Available at https://cds.climate.copernicus.eu/datasets/satellite-lai-fapar?tab=documentation

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ATBD-MGVI	Envisat's Medium Resolution Imaging Spectrometer – Algorithm Theoretical Basis Document: FAPAR and Rectified Channels over Terrestrial Surfaces, Issue 3.0, 12th May 2011

Available at https://publications.jrc.ec.europa.eu/repository/bitstream/JRC65248/atbd_mgvi_new.pdf

MOD17-UG	User's Guide GPP and NPP (MOD17A2/A3) Products NASA MODIS Land Algorithm, Version 2.0, 2nd December 2003.

Available at https://lpdaac.usgs.gov/documents/212/mod17_v5_user_guide.pdf

ATBD-MOD17	MODIS daily photosynthesis (PSN) and annual net primary production (NPP) product (MOD17) - Algorithm Theoretical Basis Document, Version 3.0, 29th April 1999.

Available at https://modis.gsfc.nasa.gov/data/atbd/atbd_mod16.pdf

MOD12Q2_User_Guide_V61	User Guide to Collection 6.1 MODIS Land Cover Dynamics (MCD12Q2) Product, 14th May 2022

Available at https://lpdaac.usgs.gov/documents/1417/MCD12Q2_User_Guide_V61.pdf

VNP22_User_Guide_V1	VIIRS Global Land Surface Phenology Product User Guide, Version 1.1, March 2020.

Available at https://lpdaac.usgs.gov/documents/637/VNP22_User_Guide_V1.pdf

CLMS_HRVPP_VR_ST-VPP	Copernicus Land Monitoring Service. High-Resolution Vegetation Phenology and Productivity (HR-VPP), Seasonal Trajectories and VPP parameters. Validation Report 2021-2024

Available at https://land.copernicus.eu/en/technical-library

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2 THE COPERNICUS LAND MONITORING SERVICE "VEGETATION & ENERGY" PRODUCTS

This chapter describes the (group of) products of the CLMS portfolio belonging to the "Vegetation & Energy" thematic. The history of the products/versions is recalled, the main characteristics of each (group of) products are described with a particular focus on those analysed during the quality assessment process. The main results of the quality evaluation of the currently disseminated products/versions are also summarized.

2.1 THE SENTINEL-3 TOP-OF-CANOPY (TOC) REFLECTANCES

The Sentinel-3 (S3) TOC reflectances are the common input for the LAI, FAPAR, FCover (Section 2.2) and DMP/GDMP/NPP/GPP (Section 2.3) calculation, for the Bidirectional Distribution Reflectance Function (BRDF) normalization to create NDVI (Section 2.4), LSP (Section 2.5), for the Burned Areas (Section 2.6) detection and for the evapotranspiration and heat fluxes (Section 2.7) retrieval. The pre-processing chain produces the S3 TOC reflectances from the original Top-Of-Atmosphere (TOA) reflectances acquired by the Ocean and Land Color Imager (OLCI) and the Sea and Land Surface Temperature Radiometer (SLSTR). The pre-processing chain consists of several steps: i) generating the S3 OLCI and SLSTR frames with cloud mask; ii) reprojecting these frames into the CGLOPS 300m grid and creating 10°x10° tiles; iii) performing the atmospheric corrections.

The cloud, cloud shadow and snow masking are based on separate schemes for OLCI and SLSTR data. The OLCI pixels are processed by the IdePix classification algorithm based on a neural network approach [ATBD_IdePix]. For SLSTR observations, the cloud masking comes from the summary cloud flag contained in the TOA reflectance data files.

The SMAC (Simplified Model for Atmospheric Correction) (Rahman and Dedieu, 1994) was chosen to perform the atmospheric corrections on OLCI and SLSTR TOA reflectances thanks to its speed, robustness and simplicity. In addition to TOA reflectances, the input data are solar and viewing angles, latitude and longitude and information characterizing the surface and atmosphere (surface elevation, pressure, content of water vapor and ozone, aerosols). The GTOPO30 (Global 30 Arc-Second Elevation) dataset is used for assigning a surface elevation to each pixel. Aerosol optical thickness (AOT) and total column of ozone are obtained from the Copernicus Atmosphere Monitoring Service (CAMS) forecast near-real time data. Surface pressure and total column of water vapor are included into the Sentinel-3 input files. Finally, the aerosol model is assumed to be continental and the surface to be Lambertian. The SMAC also allows providing an error on the TOC reflectances due to the uncertainties on aerosol model, ozone and water vapor contents, surface pressure, AOT and TOA reflectance. Moreover, due to the limitations of the SMAC, which gives degraded results for grazing zenith angles (> 65°) and high aerosol optical thickness (> 1), a quality flag is provided, on a

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pixel basis, to inform on the performances of the atmospheric correction scheme. A full description of the atmospheric corrections can be found in [CGLOPS1_ATBD_S3-AC-V1.1].

The performances of the Sentinel-3 atmospheric corrections have been evaluated following the ACIX (Atmospheric Correction Intercomparison eXercise) strategy (Doxani et al. 2018). Results, presented in [CGLOPS1_QAR_S3-AC-V1.1], exhibit that Sentinel-3 TOC reflectances are reliable and can be used with good confidence as inputs for the generation of biophysical products.

The quality assessment of the Sentinel-3 TOC reflectances, derived from OLCI and SLSTR observations, has been performed according to the methodology detailed in Section 5.1, using the in-situ reference datasets described in Section 4.1.1 and the satellite reference products presented in Section 4.2.1. The exercise demonstrates that the Sentinel-3 TOC reflectances provide reliable values at global scale and similar spatial patterns than Sentinel-2 images at local scale. The analysis results are detailed in the VR [CGLOPS1_VR_S3-TOCr300m-v2.3].

2.2 THE LAI, FAPAR, FCOVER PRODUCTS

The Version 2.0 algorithm for PROBA-V and Sentinel-3 LAI, FAPAR and FCover 300m products [CGLOPS1_ATBD_LAI[FAPAR, FCOVER]300m-V2] capitalizes on the development and validation (Fuster et al., 2020) of already existing Version 1.0 300m products, produced and delivered in Near Real Time (NRT) from January 2014 to June 2020, from PROBA-V data [CGLOPS1_ATBD_LAI[FAPAR, FCOVER]300m-V1].

The Version 2.0. of the algorithm [CGLOPS1_ATBD_LAI[FAPAR, FCOVER]300m-V2] was adapted to the new collection of PROBA-V data, Collection 2, and to the Version 2.3 of Sentinel-3 OLCI surface reflectance (Section 2.1). First, neural networks trained with Version 1.0 PROBA-V 300m products derive instantaneous estimates of LAI, FAPAR, FCover from PROBA-V and Sentinel-3 top of canopy (TOC) reflectances data. Second, product compositing at a dekadal step is performed through dedicated temporal smoothing techniques using a dissymmetric and variable length temporal window (between [-15, 0] days to [-210, +60] days), depending on the number of valid observations. Specific neural networks and compositing process are applied to evergreen broadleaf forests that are highly affected by cloud occurrence, which induces higher noise levels (cloud misdetection) and missing data. In real time mode, RT0, the compositing is achieved using only the past observations. Each time a new dekad is processed (real time estimates), the recent past values of the variables are updated. This results in successive updates of the LAI, FAPAR, FCover in the course of next 6 dekads from near real time to consolidation after six dekads. The products are delivered in near-real-time (RT0), after one and two dekads (RT1 and RT2) in the consolidation period, and after 6 dekads at the end of the consolidation period (RT6).

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The products are delivered with quality indicators including the number of valid instantaneous estimates available in the compositing window, the root mean square errors of the dekadal values compared to the instantaneous estimates in the compositing window and the length in days of the semi-periods of composition before and after the date of the product.

The quality assessment of the Sentinel-3 LAI, FAPAR, FCover V2.0 has been conducted following the procedure described in Section 5.2, using the in-situ reference datasets introduced in Section 4.1.2 and the satellite reference products presented in Section 4.2.2. The analysis demonstrates that the continuity of the 300m time series, initiated with PROBA-V V2.0 products, is ensured on a reliable manner thanks to the switch to OLCI observations. The detailed results are compiled in the validation report [CGLOPS1_VR_LAI[FAPAR,FCOVER]300m-V2.0].

2.3 THE DMP, GROSS DMP, NPP, AND GPP PRODUCTS

A simplified Monteith approach, using remote sensing data to provide the fraction of absorbed PAR-radiation (FAPAR), was first implemented by Veroustraete (1994), providing NPP estimates for European forests. This basic algorithm was consequently applied for entire Europe in the frame of the MARSOP project to produce DMP estimates on a regular basis since around 2000, with unaltered parameterization. Over the years, different versions of the DMP product arose which were distributed via separate channels to the user community. From 2004 onwards, VITO produced global DMP images derived from the 10-daily syntheses of SPOT-VEGETATION. This version was upgraded, using improved meteorological data and FAPAR inputs, and adopted by the CLMS as DMP 1km version 1 [GIOGL1_ATBD_DMP1km-V1].

Within CLMS, this original 1km DMP (version 1) has evolved in a second version at 1km, using CLMS FAPAR 1km Version 2 as input, with updated CO_2 fertilization factor, FLUXNET based biome specific light use efficiencies (LUE) as suggested by several authors (e.g., Garbulsky et al., 2010; Turner et al., 2003), and with updated autotrophic respiration factor [CGLOPS1_ATBD_DMP1km-V2]. These developments resulted in less perturbations by cloud and snow contaminated values, especially in the northern hemisphere winter, and less unreliable DMP values in regions with persistent cloud cover. The 300m DMP products derived from PROBA-V [CGLOPS1_ATBD_DMP300m-V1], from Sentinel-3 [CGLOPS1_ATBD_DMP300m-V1.1], and from a combination of both satellites [CGLOPS1_ATBD_DMP300m-V2] use the same algorithm as the DMP 1km version 2, but use FAPAR 300m V1.0 [CGLOPS1_ATBD_LAI[FAPAR, FCOVER]300m-V1], FAPAR 300m V1.1 [CGLOPS1_ATBD_LAI[FAPAR, FCOVER]300m-V1.1], and FAPAR 300m V2 [CGLOPS1_ATBD_LAI[FAPAR, FCOVER]300m-V2] as input data, respectively. The DMP 300m Version 2 products are delivered with global coverage, containing DMP, GDMP, NPP, and GPP with associated quality flag indicators (QFLAG). The quality of DMP

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is partly depending on the quality of the FAPAR (*in casu* the Copernicus Land Monitoring Service FAPAR, section 2.2), the quality of the meteorological input parameters (European Centre for Medium-Range Weather Forecasts – ECMWF data), and partly on the model.

The quality assessment of the DMP/GDMP/NPP/GPP V2 products has been conducted following the procedure described in Section 5.3, using the in-situ reference datasets introduced in Section 4.1.3 and the satellite reference products presented in Section 4.2.3. The exercise demonstrates that DMP/GDMP/NPP/GPP V2 is remarkably similar to PROBA-V DMP/GDMP V1.0 and Sentinel-3/OLCI DMP/GDMP/NPP/GPP V1.1 regarding the statistical and visual comparisons, but the completeness of V2 is higher than in previous versions. Furthermore, the temporal stability of DMP/GDMP/NPP/GPP V2 is improved compared to the combination of PROBA-V DMP/GDMP V1.0 and Sentinel-3/OLCI DMP/GDMP/NPP/GPP V1.1. The complete results are detailed in the validation report [CGLOPS1_VR_DMP-NPP300m_V2].

2.4 THE NDVI PRODUCTS

The NDVI 300m Version 3 product is derived from PROBA-V and Sentinel-3/OLCI TOC reflectances. The algorithm is described in [CGLOPS1_ATBD_NDVI300m-V3]. A BRDF normalization [CGLOPS1_ATBD_BRDFCorrection300m-V1.6] is applied on the different red and NIR spectral bands. For this purpose, red and NIR TOC reflectances are accumulated over a period of 30 days. Observations are processed sequentially by the REgularized Brdf invErsion for Land Surface reflectances (REBELS) code, yielding a BRDF corrected TOC reflectance (TOC-r) per band. These angular normalized reflectances (TOC-r) in the NIR and red bands are used to compute a BRDF corrected NDVI product. Sentinel-3/OLCI has several narrow spectral bands in red and NIR, in contrast to the preceding PROBA-V sensor having single broad red and NIR bands. Consequently, prior to NDVI calculation, a selection of Sentinel-3/OLCI red bands (Oa07 and Oa08) and Sentinel-3/OLCI NIR bands (Oa16 and Oa18) are averaged to a broad red and a broad NIR band. Additionally, a correction factor of 4.5% is applied to the PROBA-V NDVI values to minimise the bias between NDVI from both sensors.

The NDVI 300m Version 3 product is distributed with the following per-pixel information layers: (i) NDVI, (ii) uncertainty associated to the NDVI (NDVI_unc) (iii) quality flags (QFLAG), and (iv) number of recent observations used in the BRDF inversion procedure (NOBS).

NDVI 300m V3 is derived from PROBA-V Collection 2 daily (S1) TOC data and from reprocessed Sentinel-3/OLCI Surface Reflectance V2.3 data (Section 2.1). In contrast, the previous versions, NDVI 300m V1 and NDVI 300m V2, were derived from PROBA-V Collection 0 and Collection 1 10 daily (S10) products and from Sentinel-3/OLCI Surface Reflectance V1 data, respectively. In the

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latest version, i.e. NDVI 300m V3, BRDF normalization is applied to the whole time series, including the PROBA-V period.

The NDVI 300m V3 product has been evaluated following the procedure described in Section 5.4, using the satellite reference products presented in Section 4.2.4. The analysis shows that the NDVI 300m V3 displays high completeness, good stability, and large statistical consistency with the reference products. The detailed results are compiled in the quality assessment report [CGLOPS1_QAR_NDVI300m-V3].

2.5 THE LAND SURFACE PHENOLOGY PRODUCTS

The global Land Surface Phenology (LSP) 3ààm Version 2 products describe the seasonal dynamics of vegetated land surfaces inferred from satellite observations. They provide annual Vegetation Phenology and Productivity (VPP) parameters at approximately 300 m spatial resolution from 2014 onwards, based on PROBA-V and Sentinel-3 OLCI BRDF-corrected TOC reflectance time series [CGLOPS1_ATBD_BRDFCorrection300m-V1.6]. The parameters characterize the timing, magnitude, and proxies of total productivity of up to two growing season per year. The processing chain comprises the following four steps [CGLOPS1_ATBD_LSP300m-V2]:

1. Data preparation: generating nadir BRDF-corrected TOC reflectance using REBELS Adjusted Reflectance retrieval algorithm [CGLOPS1_ATBD_BRDFCorrection300m-V1.6] and collecting ECMWF ERA5 diffuse fraction of solar radiation.
2. PPI calculation: generating Plant Phenology Index (PPI) from the red and near-infrared BRDF-corrected TOC reflectances at 300 m spatial resolution and five-day time step. PPI is a physically based vegetation index developed from a simplified solution to the radiative transfer equation and is designed to respond to changes in photosynthetically-active green leaf area (Jin and Eklundh, 2014).
3. Time-series smoothing: smoothing and gap-filling the PPI time series with cubic spline functions implemented in TIMESAT (Jönsson and Eklundh, 2004), version 4.2. The fitted curves provide continuous seasonal trajectories from which the VPP parameters are derived. Cubic spline functions have high fidelity and flexibility to represent satellite measurements of plant phenology (Bolton et al. 2020; Cai et al. 2017).
4. VPP parameter extraction: extracting 13 phenology and productivity parameters for up to two growing seasons per year from the fitted seasonal trajectories. Seasonal trajectories are used internally and are not distributed as output products. The parameters are the dates of start, maximum and end of season (SOSD, MAXD, and EOSD); length of season (LENGTH); PPI values at start, maximum, and end of season (SOSV, MAXV, and EOSV); minimum PPI value (MINV); seasonal amplitude (AMPL); slopes of the green-up and green-

down periods (LSLOPE and RSLOPE); and total and seasonal productivity (TPROD and SPROD). TIMESAT determines SOSD and EOSD by applying relative thresholds 25% and 15% respectively, of the seasonal amplitude. The remaining parameters are then derived from the fitted trajectory and the detected SOSD and EOSD.

The quality of LSP Version 2 products have been evaluated by comparison has been conducted following the procedure described in Section 5.8, using the in-situ reference datasets introduced in Section 4.1.7 and the satellite reference products presented in Section 4.2.7. Results confirm that LSP v2 dataset is consistent, accurate and robust, providing marginally shorter growing season than in-situ observations. Details of the analysis and full results are available in the validation report [CGLOPS1_VR_LSP300m-V2].

2.6 THE BURNED AREAS PRODUCTS

The global Burned Areas product Version 4.0 are derived from Sentinel-3 OLCI and SLSTR TOC reflectances (section 2.1). The retrieval methodology is based on a deep learning algorithm trained with synthetic datasets generated from semi-empirical models of fire-induced spectral reflectance changes and anisotropy of surface reflectance (Padilla et al., 2025). The models are calibrated with reflectance data at active fire detections. The deep learning algorithm is used to predict burned area, on a per-pixel basis, from time series of reflectance observations [CGLOPS1_ATBD_BA300m-V4.0]. This burned area detection method is designed to deliver 1) a daily NRT product, available with each new image acquisition, and 2) a monthly non-time critical (NTC) product, with improved accuracy but available few weeks later.

The NRT and NTC Burned Areas products Version 4.0 have been validated following the procedure described in Section 5.5, using the reference products presented in Section 4.1.6. Benchmarking has been performed using the satellite products presented in Section 4.2.6. The analysis demonstrates that the Burned Areas products Version 4.0 NTC product achieves higher accuracies than other broadly used global products, while the NRT product reaches comparable accuracy levels, with the additional advantage of near-real-time availability (Padilla et al., 2025 and [CGLOPS1_VR_BA300m-V4.0]).

2.7 EVAPOTRANSPIRATION AND HEAT FLUXES

The retrieval of actual evapotranspiration (ETA) and Heat Flux (HF) products is based on two modelling frameworks. The first one is the Sen-ET framework (Guzinski et al., 2020, 2021) developed to model ETA with Copernicus data at various spatial scales and using the Two-Source Energy Balance Priestley-Taylor (TSEB-PT) ET model (Norman et al., 1995; Kustas and Norman,

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1999; Anderson et al., 2024). The second is the WaPOR framework developed by FAO through the WaPOR project (https://www.fao.org/in-action/remote-sensing-for-water-productivity/en/) and using the WaPOR-ETLook model which is based on the ETLook ETA model (Bastiaanssen et al., 2012). WaPOR-ETLook will be referred to as ETLook for the reminder of the document. Both models, although conceptually different, estimate evaporation and transpiration and use land surface temperature (LST) as one of core input forcings.

Modelling actual evapotranspiration and heat fluxes with satellite observations and TSEB-PT or ETLook models is a complex task requiring diverse set of input forcing variables. This includes shortwave optical imagery from Sentinel-3 satellites (i.e. Top-Of-Canopy (TOC) reflectance) needed to derive biophysical properties of the surface (e.g. leaf area index (LAI) and albedo), thermal infrared observations of land surface temperature (LST) from Sentinel-3 satellites which is the boundary condition for the land surface - air energy exchange and a proxy for root-zone soil moisture, weather forcings from Copernicus Atmosphere Monitoring Service, which drive (e.g. solar irradiance) and modulate (e.g. wind speed) the energy exchange between the land surface and air, and ancillary data (e.g. digital elevation models and canopy height maps) that cannot be derived from the other data sources.

Actual evapotranspiration and heat fluxes have been validated following the procedure described in Section 5.9 using the in-situ reference datasets introduced in Section 4.1.3 and the satellite reference products presented in Section 4.2.8. Analysis shows that the CLMS ETA and HF the quality of the current state of the art in remote sensed-based ET modelling. Detailed results are presented in the validation report [CGLOPS1_VR_ETA300m-V1].

2.8 THE LAND SURFACE TEMPERATURE PRODUCTS

The Land Surface Temperature (LST) product is retrieved from a constellation of geostationary satellites, operated by different agencies (Freitas et al., 2013). The original product used data from Meteosat Second Generation (MSG), Geostationary Operational Environmental Satellite (GOES) and the Multi-Function Transport Satellite (MTSAT). Since then, MTSAT was replaced by a third-generation platform in November 2015, the Himawari-8 satellite; and the same happened in the GOES East coverage, with GOES-16 entering in operations in December 2017. While the MSG nominal 0º mission remained unchanged, the first MSG (Meteosat-8) was moved to the 41.5° E longitude, ensuring the Indian Ocean Data Coverage (IODC) mission. The latter was taken over by the second MSG (Meteosat-9) in June 2022, at the target longitude of 45.5º E. The LST V3.0 includes observations from GOES-16/19 (transition in April 2025), Himawari-8/9 (transition in December 2022), MSG prime at 0º, and MSG-IODC missions. LST is retrieved for each satellite mission every hour for all the clear-sky pixels, using the Generalized Split-Windows (GSW) algorithm. This

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algorithm has been used in operational retrievals, including LSA-SAF LST products (Trigo et al., 2008), and is based on a formulation similar to the approach proposed by Wan and Dozier (1996) for AVHRR and MODIS. It combines brightness temperatures from two thermal infrared channels with ancillary data, including surface emissivity, total column water vapour from Numerical Weather Prediction fields and a cloud mask. In V3.0, updates were introduced to the input datasets and processing chain, including: 1) enhanced cloud screening based on Nowcasting-SAF information (NWC-SAF); 2) revised snow-cover treatment using a gap-filling approach that combines snow fraction information from the CLMS Snow Cover Extent (SCE) 1 km product (Metsämäki et al., 2012, 2015) with the Interactive Multisensor (IMS) Daily Northen hemisphere Snow and Ice Analysis 4 km product (U.S. National Ice Center, 2008); 3) updated emissivity inputs (for GOES and Himawari); 4) spatial resolution upscale to 0.025º (approximately 3 km at the equator); 5) overall algorithms refactoring; and 6) the consistent application of these updated inputs since 2018 to present. The hourly product is distributed with an associated uncertainty, derived by propagating the uncertainties of the input datasets and the retrieval algorithm (Freitas et al, 2010). While the LST retrieved from the MSG satellites is directly ingested from the LSA-SAF data files, the GOES and Himawari retrievals rely on an optimized algorithm calibration strategy (Martins et al., 2016). The individual estimates from each satellite mission are then combined into a global product on a regular 0.025º grid, using the algorithm described in Freitas et al (2013) and in the corresponding ATBD [CGLOPS1_ATBD_LST-V3.0].

To facilitate the integration of LST products with other CGLOPS variables in applications such as agrometeorological monitoring, a 10-day composite LST product is also available. This product provides a characterization of the LST daily cycle for each pixel over each 10-day compositing period (LST10-DC). Additionally, a 10-day synthesis and Thermal Condition Index (TCI) products are produced and distributed as part of the 10-daily group of products (LST10-SYN, LST10-TCI, respectively). The global LST hourly maps are used to estimate the main parameters describing the diurnal cycle for each pixel, including maximum, minimum, and median temperatures, together with the fraction of valid pixels used in the calculation. These parameters are estimated for each 10-day period, with the corresponding statistics being then derived and compared to a multi-year climatology to produce the TCI. Further details are given in the LST10 ATBD [CGLOPS1_ATBD_LST10-V3.0].

The LST V3.0 and LST10 V3.0 family of products have been evaluated according to the validation methodology described in Section 5.6, using the in-situ reference datasets presented in Section 4.1.4, as well as the satellite and reanalysis datasets introduced in Section 4.2.5 and Section 4.3.2. The assessment includes comparisons with ground-based observations and independent datasets to evaluate the accuracy, temporal stability, and spatial consistency of the LST V3.0 family of products. The detailed results are compiled in the corresponding validation report [CGLOPS1_VR_LST-LST10-V3.0].

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2.9 THE SSM AND SWI PRODUCTS

Soil moisture is a key parameter in numerous environmental studies including hydrology, meteorology, and agriculture. Surface Soil Moisture (SSM) is the moisture content in the upper few centimeters of the soil, and is directly accessible to spaceborne radar sensors, as Sentinel-1 or Metop Advanced Scatterometer (ASCAT). Soil moisture in plant root zone can be estimated by an infiltration model using information on surface soil moisture and soil characteristics, estimating the so-called Soil Water Index (SWI) Wagner et al., 1999; Albergel et al., 2008).

The CLMS Surface Soil Moisture Version 2.0 product, named SSM1km, describes soil moisture of the soil's topmost 5cm on a 1km (1°/112) spatial sampling, and is operationally provided with global coverage. It is derived from microwave radar data observed by the Sentinel-1 Synthetic Aperture Radar (SAR) satellite sensors, and includes associated uncertainty estimates. Soil moisture is estimated from the backscatter observations by inverting a zeroth-order radiative transfer model RT0 (Wagner et al., 2022, Quast et al., 2023 and Raml et al., 2026), which models vegetation dynamics using auxiliary Leaf Area Index (LAI) data from the CLMS global LAI 300m product. The RT0 model represents an advanced version of the TU Wien Change Detection model used in earlier product versions (Bauer-Marschallinger et al., 2019), for which evaluations detected positive biases during the growing season. The explicit vegetation modelling of RT0 addresses these biases. In addition, radar pixels with no sensitivity to soil moisture, as caused for instance by subsurface scattering over arid regions, are masked at fine scale before aggregation to 1km. Initial experiments show that this masking substantially improves the retrievals over arid and semi-arid regions (Raml et al., 2026).

The Soil Water index (SWI) product of the CLMS provides global daily information about moisture conditions in different soil depths, and is provided within CLMS in two resolutions and production windows:

1. First, the globally available SWI 0.1° version 4.0 product ingests near-real-time data of the H SAF Meteop ASCAT surface soil moisture products as input data. The SWI10 product aggregates the SWI data over ten days intervals to make combination with other CGLS products easier. The SWI time series product (SWI-TS) contains the same data as the SWI product but is reformatted for easier time series access. All SWI datasets include a Quality Flag (QFLAG) indicating the availability of SSM measurements for the SWI calculations.
2. Second, the SWI1km version 3.0 product combines radar soil moisture estimates from Metop ASCAT and Sentinel-1 and aims to provide SWI estimates with a high spatio-temporal resolution. The SWI1km is built upon data streams existing in the CLMS for the SWI 0.1° and SSM1km products, following the method of Bauer-Marschallinger et al., 2018. In

particular, it uses the high-resolution SAR surface soil moisture (SSM) data generated by the 2014-launched Sentinel-1 mission and combines it with the ASCAT scatterometer-based SSM data. The resulting daily product provides a 1km (1°/112) spatial sampling, and achieves both, high temporal and spatial resolution. It furthermore provides full coverage over operated areas, allowing hydrological applications of radar data on a new level. The CLMS SWI1km product is operationally provided with global coverage, and also provides QFLAG and SSF information. The product quality is assessed in the accompanying validation report.

SSM and SWI are evaluated according to the procedure described in Section 5.7 using the in-situ reference dataset presented in 4.1.5 as well as the models output datasets introduced in Section 4.3.1. Complete results are included in the accompanying validation reports [CGLOPS1_VR_SSM1km-V2, CGLOPS1_VR_SWI1km-V3, CGLOPS1_VR_SWI-V4].

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3 OVERALL QUALITY ASSURANCE STRATEGY

The activities performed to check the quality of the CGLOPS "Vegetation and Energy" products are split in two categories:

- A robust scientific validation following, as much as possible and when existing, the guidelines, protocols and metrics defined by the Land Product Validation (LPV) group of the Committee on Earth Observation Satellites (CEOS) for the validation of satellite-derived land products (https://lpvs.gsfc.nasa.gov/).
- The operational quality monitoring carried out at production centres, each time a new file is generated, and scientific quality monitoring are described into the System Quality Assurance Document (SQAD).

The overall validation strategy of CGLOPS biophysical products relies on three major components:

1. Completeness corresponding to the frequency and continuity of products.
2. Indirect validation: it consists of inter-comparisons with existing other satellite-derived products. Those latter are described in Section 4.2. This includes the evaluation of their temporal and spatial consistency.
3. Direct validation: it corresponds to the comparison of temporally and spatially concurrent satellite-derived products with in-situ reference datasets. Those latter are described in Section 4.1. The direct validation can be limited by the small number of reference datasets.

These generic guidelines are adapted to fit the specificities of each (group of) CGLOPS products, as detailed in Chapter 5.

The robust scientific validation is carried out for each new product to be introduced into the portfolio and each new version of existing products. Results are gathered in a Validation Report (VR) available online in the CLMS web portal (https://land.copernicus.eu/en). Each year, the validation and quality assessment reports are updated with the results of the scientific quality monitoring in order to retain the validation stage 4 as defined by the CEOS/LPV (https://lpvs.gsfc.nasa.gov/).

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4 REFERENCE DATASETS

This chapter describes the external datasets used as reference in the quality assessment and validation exercises. It includes the in-situ measurements used for the direct validation, the existing similar satellite products and output models used for intercomparison or benchmarking.

4.1 IN-SITU REFERENCE DATASETS

4.1.1 TOC Reflectances

- Radiometric Calibration Network (RadCalNet)

RadCalNet (Bouvet et al., 2019) is an initiative of the Working Group on Calibration and Validation (WGCV) of the CEOS. RadCalNet is a network of sites that can be used to compare different satellite sensors to each other and to a common reference. For each site, measurements of nadir-viewing TOC reflectance along with atmospheric parameters (such as surface pressure, columnar water vapor, columnar ozone, aerosol optical depth, and the Angstrom coefficient to define aerosol sizes) are provided. Simulation to TOA reflectance is performed through the central RadCalNet Processing for all sites, ensuring that the in-situ data from those sites are SI-traceable (SI – from Système International (d'unités)). TOA reflectance and TOC reflectance data are provided for a nadir view at 30 min intervals from 9 am to 3 pm local standard time for a given site, along with their associated uncertainties (https://www.radcalnet.org/). The spectral range of RadCalNET measurements is from 380 nm to 2500 nm, with 10 nm of spectral sampling interval.

TOC reflectances from RadCalNet should be compared with Sentinel-3 TOC reflectances acquired under similar illumination and viewing configurations. The Spectral Response Function (SRF) coefficients of Sentinel-3 sensors are used to perform the spectral harmonization.

- ESA Fiducial Reference Measurements

ESA has created a program of work to address the concept of Fiducial Reference Measurements, a suite of independent, fully characterized, and traceable ground measurements that follow the guidelines outlined by the Quality Assurance framework for Earth Observation (QA4EO).

Ground measurements from Las Tiesas, Barrax, Spain are available (https://frm4veg.org/). The validation campaign took place on the 2nd August 2018 (Origo et al., 2020). Measurements were made over the alfalfa and bare soil surface types. The procedure utilized several key instruments in order to collect the appropriate data for the validation: a spectrometer, Spectralon panel, and sunphotometer. The spectrometer has a 400 nm–2500 nm wavelength range with a variable bandwidth of 3 nm at 700 nm and 10 nm at 1400/2100 nm. For each of the cover types, 10 sample

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points were selected over a 200 × 200 m^2 area, where it was ensured that every point was, at least, 50 m from the cover type. The measurement window was selected with the aim of conducting measurements no more than 30 min either side of the Sentinel-2A overpass, and these measurements could be useful for validating Sentinel-3 as well. This was done to minimize the change in solar zenith angle (θ_s) during the measurement window and thereby minimize any BRDF effects.

- HYPERNETS

The HYPERNETS project (www.hypernets.eu) (Ruddick et al., 2024) aims to ensure that high-quality hyperspectral surface reflectance measurements are available to support the validation of optical Visible and Near InfraRed (VNIR) and SWIR Copernicus products. Therefore, it established a new autonomous hyperspectral spectroradiometer (HYPSTAR® - www.hypstar.eu) dedicated to land and water surface reflectance validation with instrument-pointing capabilities.

HYPERNETS dataset corresponds to quality-checked L2B surface reflectance data products measured with HYPSTAR®-XR sensors every 30 minutes from different set of geometries (full zenith/azimuth coverage, up to 60° from nadir). The HYPSTAR®-XR (eXtended Range) instruments consist of a VNIR sensor and a SWIR sensor. They autonomously collect data between 380-1680 nm at various viewing geometries and send them to a central server for quality control and processing. The VNIR sensor spans 1330 channels between 380 and 1000 nm and the SWIR sensor has 220 channels between 1000 and 1680nm with. The HYPSTAR®-XR field of view is 5º in radiance and 180º in irradiance (Ruddick et al., 2024).

HYPERNETS complements the direct validation of Sentinel-3 TOC reflectances using the RadCalNet network in terms of angular configuration, since RadCalNet provides only nadir-viewing data. Comparisons between satellite acquisitions and in-situ measurements should be performed under similar illumination and viewing configurations. The SRF coefficients of Sentinel-3 [S3-TN-ESA-OL-660, S3-SLSTR-TG] are used to perform the spectral harmonization.

4.1.2 LAI, FAPAR, FCover

- The CEOS WGCV LPV DIRECT 2.1 dataset

The CEOS LPV DIRECT version 2.1 database (Camacho et al., 2024b) represents a major international effort to provide harmonized and suitable reference data for the validation of LAI, FAPAR and FCOVER satellite products. DIRECT 2.1 is endorsed by CEOS and hosted on the CEOS Cal/Val portal (https://calvalportal.ceos.org/lpv-direct-v2.1). It compiles ground reference upscaled values over a 3 km x 3 km area compliant with CEOS LPV guidelines.

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Ground measurements acquired over elementary sampling units (ESUs) from multiple indirect techniques were upscaled using high spatial resolution imagery following CEOS LPV protocols, ensuring proper representation of site spatial heterogeneity. The first version (DIRECT) included measurements from several international activities including Validation of Land European Remote sensing Instruments (VALERI), BigFoot, Southern African Regional Science Initiative – 2000 (SAFARI-2000), Canada Centre for Remote Sensing (CCRS), Boston University and ESA campaigns compiled by Garrigues et al. (2008) and later ingested in the CEOS OLIVE tool (Weiss et al., 2014) for accuracy assessment. The database was subsequently revised to remove sites lacking understory measurements, which had led to misinterpretation of earlier validation results (Camacho et al., 2013) and, after that, it was expanded with the inclusion of sites from the FP7 Implementation of Multi-scale Agricultural Indicators Exploiting Sentinels (ImagineS) project (Camacho et al., 2021). CEOS LPV DIRECT 2.1 is the latest update including 44 new sites from China (Fang et al., 2019; Song et al., 2021) and 2 sites from ESA FRM4Veg (Camacho et al., 2024a; Brown et al., 2021), resulting in a total of 176 sites worldwide across seven major biomes. The database contains 189 effective LAI values, 280 LAI, 128 FAPAR and 122 FCOVER values covering the period from 2000 to 2021.

- CLMS Ground-Based Observations for Validation (GBOV) service Land Products

As part of the Copernicus Global Land Service, the Ground-Based Observations for Validation (GBOV) service (https://gbov.land.copernicus.eu/) aims at facilitating the use of observations from operational ground-based monitoring networks and their comparison to Earth Observation products. In case of LAI and FAPAR, the GBOV service performs the implementation and maintenance of a database for the distribution of reference measurements (RMs) and the corresponding Land Products (LPs) (i.e., upscaled maps). Currently, GBOV provides multi-temporal Land Products over 27 sites.

The current version (V3) of GBOV LP algorithm [ATBD-GBOV-LP3-LP4-LP5] takes as input the Reference Measurements (RMs) collected over a given site, in addition to a series of high spatial resolution images. Calibration functions are then derived between RM and Radiative Transfer Model (RTM)-based retrievals, enabling high spatial resolution maps of each RM to be produced.

The use of calibrated RTM-based retrievals (GBOV V3) as opposed to vegetation index-based multi temporal transfer functions in previous version (GBOV V2) enables the impact of non-canopy factors that perturb the vegetation index-biophysical variable relationship to be reduced. For example, as viewing and illumination angles are an explicit input, seasonal variations in sun-sensor geometry can be better accounted for, whilst the variety of soil spectra used in the RTM simulations helps reduce the impact of the soil background (Brown et al., 2021). To maintain computational efficiency, a hybrid method using artificial neural networks trained with RTM simulations was selected as opposed to a pure inversion approach.

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It is important to emphasize that the use of a generic transfer function approach in GBOV—as opposed to the site-dependent transfer functions recommended and widely used in the CEOS LPV (e.g., DIRECT 2.1 database (Camacho et al., 2024b))—raises significant concerns about potential biases at certain sites. Consequently, before using the GBOV dataset, a quality control is performed, which consists of assessing the consistency of RM with LPs over visually homogeneous forest sites, where direct comparison of each RM with LP at 300 m spatial resolution can be made without expecting large systematic deviations due to the spatial heterogeneity of the sites.

- EOLAB FRM-Compliant campaigns

EOLAB has recently conducted independent field campaigns at different locations to collect Fiducial Reference Measurements (FRM) for the validation of LAI, FAPAR and FCOVER satellite products. The EOLAB FRM dataset is available in Camacho et al., 2026. Observations were derived from indirect optical methods, including digital hemispherical photography, LAI-2200 Plant Canopy Analyzer, and AccuPAR ceptometer, following ESA FRM4Veg protocols for both measurement and uncertainty estimation (Brown et al., 2021; Camacho et al., 2024a). The current version comprises 189 sampling units across several field campaigns over three cropland sites in Spain and one site in the Amazon Basin (Peru): three field campaigns (May, July and November 2017) in Moncada (Spain), one campaign (18–21 July 2022) in Barrax (Spain), one field campaign (1 August 2024) in Nules (Spain), and one field campaign (21 to 26 September 2024) in Chontachaka (Peru). The dataset will be updated as new campaigns become available.

The upscaling of the FRM measurements was performed in agreement with CEOS LPV recommendations, using Sentinel-2A Multispectral Instrument (MSI) surface reflectance (L2A) data, to generate FRM-based high-spatial resolution reference maps for the validation of hectometric satellite products. An area of 5 km by 5 km was typically selected. The uncertainties associated with the FRM measurements were propagated through the transfer functions to derive an uncertainty layer for each variable in agreement with FRM4Veg procedures (Camacho et al., 2024a).

4.1.3 Flux tower measurements

At a number of individual nodes (sites) spread throughout the globe, micrometeorological towers are located that use eddy covariance measurements to estimate a wide range of ecosystem variables like exchanges of carbon dioxide, water vapour, and energy between the biosphere and atmosphere. A number of different regional networks exist, such as the European Integrated Carbon Observation System (ICOS), Ameriflux for the US, and Ozflux for Australia that coordinate the collection, processing, and distribution of the Eddy Covariance (EC) data records. At global scale, FLUXNET

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is a network combining the regional networks. The latest global data release however dates from 2015.

The flux towers measure Net Ecosystem Exchange (NEE), a measure for the total vertical fluxes of carbon between atmosphere and ecosystem. Reichstein et al. (2005) provides a "flux-partitioning algorithm" methodology to separate the observed NEE in the Gross Primary Productivity (GPP) and Ecosystem Respiration (Reco). The post-processing of flux data into variables such as GPP requires the necessary tools and expertise and hence should be performed by EC experts. Flux networks such as ICOS (https://data.icos-cp.eu/portal) and Ameriflux (https://ameriflux.lbl.gov) safeguard the quality and processing of flux data for multiple sites and are hence the best available data sources for the CLMS GPP quality monitoring.

The quality of the flux tower GPP data depends on the original measurements of NEE and on the conversion of NEE into GPP. The NEE data of each tower is checked for its quality, and gaps in the time series are filled. The method of Reichstein et al. (2005) relies on a short-term relationship between the respiration coefficient and temperature which depends on the noisiness of the eddy covariance data and the range of temperatures encompassed during the short period. On sites with very stable temperatures or noisy data the method can fail to build a reliable relationship, and this can lead to unrealistic estimates of the respiration factor and thus the derived GPP.

The direct validation of the ETA and HF products also relies on in-situ measurements of surface heat fluxes at eddy covariance stations. In gathering the in-situ reference dataset, the preferred data sources were those corrected to ensure the energy balance closure (EBC) by assuming that the initial Bowen ratio was correct. Datasets with EBC corrected data (following the ONEFlux processing pipeline (Pastorello, et al., 2020)) were fetched from the ICOS (Level 2 and Warm Winter 2020 datasets), the AmeriFlux, and the OzFlux (https://data.ozflux.org.au/home.jspx) networks. The reference dataset was enriched with data from other networks that did not implement any form of EBC correction. Those networks were: the JapanFlux dataset (https://www.japanflux.org), the BASE dataset of AmeriFlux , the European Flux Database Cluster (EFDC; https://www.europe-fluxdata.eu/), the AsiaFlux (https://asiaflux.nies.go.jp/), the AMMA-CATCH (https://www.amma-catch.org/) and the South African Environmental Observation Network (SAEON; https://observationsmonitor.saeon.ac.za).

4.1.4 Land Surface Temperature

There are few environmental monitoring networks, whose stations standard specifications include radiation measurements that are useful for LST estimation. In the VR of LST V3.0 [CGLOPS1_VR_LST-LST10-V3.0], validation is based on the most recent GBOV reference

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measurements for LST datasets (RM-9). The GBOV service (Section 4.1.2) provides LST estimates over a broad range of surface conditions (Bai et al., 2019). In particular, RM-9 datasets are usually estimated from near-surface measurements of upwelling and downwelling TIR radiances (RM-2) based on land surface emissivity (RM-8), recorded by available in-situ radiometers or pyrgeometers, as well as from other meteorological parameters [GBOV-ATBD-RM2-RM8-RM9_LST]. Since its release in September 2023, version 2.2 of RM-9 datasets has been used to evaluate the CLMS LST family of products, with a total of 59 sites being currently available from the GBOV portal.

Given the high spatial and temporal variability of LST, direct validation against in-situ measurements is challenging. Therefore, an initial data curation is performed to select only robust GBOV sites suitable for satellite LST validation. This pre-selection includes a comprehensive quality-control and data availability checks. Specifically, this selection follows four main criteria: 1) at least one full year of valid observations within the evaluation period; (2 location within the field of view of the corresponding geostationary component; (3) no instrumentation issues affecting local measurements; and (4) consistency with surface representativeness at the satellite pixel scale. The latter two criteria are assessed through a detailed inspection of raw in-situ time-series and examination of the station surrounding environment using high-resolution orography, land cover maps, and Landsat-based LST retrievals, implemented via the Google Earth Engine algorithm developed by Ermida et al. (2020). The selected sites are geographically distributed over a wide range of land cover types, vegetation conditions, elevations and climate zones. Such diversity ensures a robust assessment of the CLMS LST V3.0 dataset under varying surface and atmospheric conditions.

4.1.5 Soil Moisture

The ISMN (International Soil Moisture Network; Dorigo et al., 2011) is an international cooperation to establish and maintain a global in-situ soil moisture database. This database is an essential means of the geoscientific community for validating and improving global satellite observations and land surface models. The ISMN (https://ismn.earth/en/) provides a harmonized repository of in-situ soil moisture observations, which makes it feasible to use in-situ data from a wide array of networks. Paulik et al. (2014) carried out already an evaluation of the ASCAT SWI at a selection of the ISMN locations.

With 2024, the ISMN contains data of 80 networks and more than 3100 stations located all around the globe, with highest density over Europe and North America. The time period spanned by the entire database runs from 1952 until the present, although most datasets have originated during the last two decades. The database is continuously expanding, which means that both the number of stations and the time period covered by the existing stations are still growing. Hence, it is becoming

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an increasingly important resource for validating and improving satellite-derived soil moisture products and studying climate related trends. As the ISMN is animated by the scientific community itself, potential networks are invited to enrich the collection by sharing its in-situ soil moisture data.

The ISMN employs an automated quality control system for its Soil Moisture (SM) measurements, which flags values exceeding a certain threshold and checking the validity of SM variations in relation to changes in soil temperature and precipitation (Dorigo et al., 2013). A second flagging procedure analyses the shape of the SM time series to detect outliers, positive and negative breaks, saturation of the signal, and unresponsive sensors. Dorigo et al. (2021) reported on the recent developments within the ISMN and provide a detailed documentation on the network update activities and data quality curation. This includes an ongoing activity on quality curation of historic time series that aims for most consistent and reliable in-situ data. Consequently, the available time series data at ISMN may be altered over time in terms of flagging and value density. To ensure a long-term funding, the ISMN operations were transferred by the end of 2022 to the International Center for Water Resources and Global Change (ICWRGC) and the German Federal Institute of Hydrology (BfG), and both new hosting intuitions are dedicated to continue ongoing activities on network integration and quality check.

4.1.6 Burned Areas

The validation analysis is based on reference dataset generated by the Copernicus Climate Change Service (C3S) and is available on the e-cienciaDatos repository (https://doi.org/10.21950/BBQQU7; Franquesa et al., 2020). The methods to generate the reference data, the sampling design, and the accuracy inference formulae are detailed in Franquesa et al. (2022).

C3S BA reference data is based on supervised classifications of consecutive pairs of Landsat images and accuracy inferences are based on stratified combined ratio estimator (Cochran, 1977), taking into account the size of the sampling units.

The sampling units are defined spatially also by the Thiessen Scene Area (TSA), but temporally by calendar years, forming sampling units ‘long’ in time, somewhat similar as in the CCI Fire Disturbance Product Validation Report [Fire_cci_D4.1.1_PVR_v2.1]. The accuracy estimated with long units is expected to be higher than with short units (made from single image pairs) (Franquesa et al., 2022; [Fire_cci_D4.1.1_PVR_v2.1]). The long units allow for larger temporal overlaps between product and reference datasets what decreases the impact of temporal burn detection errors on the confusion matrix. As in Franquesa et al. (2022), products’ no-data was considered as unburned area when generating the error matrices. This is a way to take into account and ‘penalize’ products that are unable to provide estimates (no-data) over large areas. The stratification of the sampling

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design was based on (1) calendar years, (2) biomes derived from the Ecoregions 2017 map (Dinerstein et al., 2017), and (3) low and high fire activity regions. For the last stratification level, each year and biome were split in two according its 80th percentile of BA. The annual sample size (~100 TSAs, 105 for year 2019) was allocated across strata proportionally to stratum size (in number of sampling units) times the square root of BA, as recommended by Padilla et al. (2017).

4.1.7 Phenology metrics

- PEP725

The Pan European Phenology Project (PEP725, http://www.pep725.eu/) (Templ et al., 2018) has the objective to establish an open access database of plant phenology for science, education and research. The database presently holds more than 16 million records, about 97 growing stages and 338 plant species (including cultivars). The phenophases that are taking into account in the comparison with SOSD and EOSD are summarized in Table 1.

Table 1: Phases used for linking CLMS LSP SOSD and EOSD with PEP725 phenology observations.

PEP725 Phase_id	Description	Short name	Metrics
10	Gramineae: Fist true leaf emerged from coleoptile, Dicotyledons and Monocotyledons: Cotyledons completely unfolded, Perennial plants: First leaves separated	Leaf emerged	SOSD
11	First true leaf, leaf pair or whorl unfolded, Perennial plants: First leaved unfolded	Leaf unfolded	SOSD
223	Leaf unfolding (≥50%)	Leaf unfolded 50%	SOSD
95	50% of leaves fallen	Leaf fallen 50%	EOSD
205	Autumnal coloration ≥50%	Coloration 50%	EOSD

- TEMPO

TEMPO (https://tempo.pheno.fr/) (Maury et al., 2023) is the French national network of observatories whose objective is to answer how climate change influences in the seasonal rhythms of

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living organism and productivity. TEMPO counts with 10 observatories and more than 95 partners (research units, technical and public institutions, etc.).

Phenological stages are recorded following the *Biologische Bundesanstalt, Bundessortenamt und CHemische Industrie* reference system, which describes annual crops and fruit trees phenological stages. The phenological stages used as potential start of season are “Germination / sprouting / bud development” and “Leaf development (main shoot)” and the stage used as potential end of season is “Beginning of dormancy, die off of leaves, stem”.

- NEON

The National Science Foundation’s National Ecological Observatory Network (NEON, https://www.neonscience.org/) is a continental-scale observation facility designed to collect long-term open access ecological data to better understand ecosystem changes. Neon provides free and open data that characterize plants, animals, soil, nutrients, freshwater and the atmosphere.

The NEON Plant Phenology observations dataset [NEON.DOC.014040] contains quality-controlled native sampling resolution data from in-situ observations of plant leaf development and reproductive phenophases. Phenophase status and intensities follow those of the USA National Phenology Network (USA-NPN). The status and intensity data are reported per phenophase, individual or patch for every sampled day. The dataset (DP1.10055.001) counts with more than615 species and has 47 sites and typically 2 different plots per site.

The phenophases used as potential start of season are "breaking leaf/needle buds” and “emerging leaves/needles,” and the phenophases used as end of season are "colored leaves/needles” and “falling leaves/needles”.

- Ecosystem Phenology Camera (PhenoCam V3)

The Ecosystem Phenology Camera (PhenoCam, https://phenocam.nau.edu/webcam/) is a network created to provide automated near-surface remote sensing of canopy phenology using high-resolution digital cameras. Images of the cameras are uploaded to PhenoCam server every half hour, extracting quantitative colour information from each picture. Canopy greenness indices provide information about the amount of foliage present and its colour. The image analysis of these cameras provides an objective means by which canopy phenology can be monitored and quantified.

The PhenoCam standard index is the green chromatic coordinate (GCC), which is the relative brightness of the green channel of cameras normalized against the overall brightness of the red, green and blue channels. From GCC time series, PhenoCam also provides an estimation of the green-up and green-down transition dates, which correspond with the start and end of the growing season. To

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estimate the transition dates, first the maximum and minimum values of greenness are estimated. Then, the 10%, 25% and 50% of the greenness curve range is computed as transition dates. PhenoCam V3 (pre-release) counts with a total of 1436 ROI's masks and data from 2000 to 2025.

4.2 SATELLITE REFERENCE PRODUCTS

4.2.1 TOC reflectances

- Sentinel-2 L2A

The Copernicus Sentinel-2 mission comprises a constellation of two polar-orbiting satellites placed in the same sun-synchronous orbit, phased at 180° to each other. It aims at monitoring variability in land surface conditions, and its wide swath width (290 km) and high revisit time (10 days at the equator with one satellite, and 5 days with 2 satellites under cloud-free conditions which results in 2-3 days at mid-latitudes) support monitoring of Earth's surface changes. The coverage limits are from between latitudes 56° south and 84° north. The main instrument of the Sentinel-2 mission is the Multi-Spectral Instrument (MSI).

Sentinel-2 L2A (level-2A) TOC reflectance products [S2-PDGS-MPC-ATBD-L2A] are generated with Sen2Cor processor (Main-Knorn et al., 2017), which is applied to granules of TOA Level-1C orthorectified reflectance products. The processing starts with the cloud detection and scene classification followed by the retrieval of the AOT and the water vapour content from the Level-1C image. The final step is the TOA to TOC conversion. The atmospheric correction is performed using a set of Look-Up tables generated via libRadtran (Mayer & Kylling, 2005). Baseline processing is the rural/continental aerosol type. More Look-Up tables enable processing with the choice between mid-latitude winter and mid-latitude summer atmosphere, rural and maritime aerosol and ozone content according to the scene geographic location and climatology. AOT is derived at 550nm using the Dense Dark Vegetation algorithm (Kaufman & Sendra, 1988), based on the correlation between reflectance in SWIR and visible bands (red and blue). Water vapour retrieval over land is performed with the Atmospheric pre-corrected differential absorption algorithm (Schläpfer et al., 1998). Level-2A outputs are the surface (or TOC) reflectance images which are provided at different spatial resolutions (60 m, 20 m and 10 m) together with AOT and water vapour maps. The internal scene classification image used for atmospheric correction is also provided together with Quality Indicators for cloud and snow probabilities. The quality status of Sentinel-2 L2A products is regularly reported by the Sentinel-2 Mission Performance Center (https://sentiwiki.copernicus.eu/web/s2-documents).

- ESA Sentinel-3 Level-2 SYN (S3_SY_2_SYN) product

The ESA Sentinel-3 Level-2 SYN product (S3_SY_2_SYN) [OMPC.ACR.HBK.003] has a resolution of 300 m and is mapped on the same regular grid than OLCI. SYN products are created

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from a combination of OLCI and SLSTR Level-1B products (including both nadir and oblique views). The Synergy swath is then corresponding to the common area between OLCI and SLSTR swaths, restricted to the daylight part of the Sentinel-3 orbit (i.e., where the sun zenith angle at satellite ground track is lower than 80°).

S3_SY_2_SYN is produced by the Synergy Level-1/2 Surface Directional Reflectance (SDR) software and contains surface reflectance (S3_SY_2_SYN) and aerosol (S3_SY_2_AOD) parameters over land. Level-2 SDR variables of S3_SY_2_SYN correspond to atmospherically corrected Surface Directional Reflectance (also called Lambert Equivalent Reflectance (LER) or Bidirectional Reflectance Factor (BRF)). They are provided with their associated error estimates for the sun-reflective SLSTR channels (S1 to S6 for both nadir and oblique views except S4 dedicated to cloud detection) and for all OLCI channels, except for the oxygen absorption bands Oa13, Oa14, Oa15, and the water vapor bands Oa19 and Oa20.

The core algorithm of this Level-2 SYN processing module is an aerosol retrieval based on an approach combining dark object and multiple view-angle methods derived from an existing work for Medium Resolution Imaging Spectrometer (MERIS) and AATSR [S3-L2-AOD-SYN-ATBD]. For a given atmospheric aerosol model (i.e., continental aerosol model) and a parameterized Aerosol Optical Depth computed at 550 nm, a set of surface reflectances is derived from OLCI/SLSTR data. The derived Error metric is then minimized with regards to angular and spectral constraints, providing the realistic value of Aerosol Optical Depth. It should be noted that the latest Level-2 SYN processing baseline includes several evolutions/corrections (SYN product notice [S3.PN.SYN-L2.09]). One of the most important evolutions is that the SLSTR calibration factors defined by SLSTR experts and the S3A OLCI offset defined during the tandem mission (2% reduced brightness for all bands (Lamquin et al., 2020), except Oa21 for which a 6% offset has been found) have been included as configuration parameters and applied at the beginning of SYN processing.

The quality assessment of S3_SY_2_SYN SDR products [OMPC.ACR.HBK.003] was performed by indirect validation against the normalized surface reflectance measurements observed from MODIS (MCD43A4.061, a combined Terra and Aqua Product) at 500 m spatial resolution and VIIRS (VNP43MA4.001) at 1 km resolution. Several regional inter-comparisons were performed, and S3_SY_2_SYN SDR reasonably compared to that of MODIS for the pair of closest bands (b01 vs Oa08), (b02 vs Oa17), (b03 vs Oa04) and (b04 vs Oa06), with R^2 values higher than 0.8. Similar agreement was when comparing VIIRS band M7 and OLCI Oa17.

4.2.2 LAI, FAPAR, FCover

- LSA SAF EPS/AVHRR Vegetation Parameters (VEGA)

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The main purpose of the Land Surface Analysis (LSA) of the Satellite Application Facility (SAF) is to take full advantage of remotely sensed data, particularly those available from EUMETSAT sensors, to measure land surface variables, which find primarily applications in meteorology (http://lsa-saf.eumetsat.int). The EUMETSAT Polar System (EPS) is Europe's first polar orbiting operational meteorological satellite program and the European contribution to a joint polar system with the United States. EUMETSAT has the operational responsibility for the "morning orbit" with Meteorological Operational (MetOp) satellites, the first of which was successfully launched on October 19, 2006. Despite the wide range of sensors on-board MetOp (http://www.eumetsat.int/), most LSA SAF parameters make use of the Advanced Very High-Resolution Radiometer (AVHRR) and, to a lesser extent, of the Advanced Scatterometer (ASCAT). The suite of LSA SAF EPS-AVHRR vegetation (VEGA) version 1.0 global products is the leaf area index (ETLAI), the fraction of photosynthetically active radiation absorbed by vegetation (ETFAPAR) and the fractional vegetation cover (ETFVC). The algorithm [ATBD-EPSVEGA] for retrieving FVC, LAI and FAPAR relies on the inversion of the PROSAIL radiative transfer model, which results from coupling the PROSPECT (Jacquemoud & Baret, 1990) and the SAIL (Verhoef, 1984) models with a family of multi-output machine learning retrieval methods (García-Haro et al., 2018).

The validation report of the EPS VEGA products [VR-EPSVEGA] demonstrated an overall good quality for most of the criteria under evaluation. EPS VEGA showed good overall statistical results compared to references CGLOPS PROBA-V 1km V1 and MODIS Collection 6 (see below) over LANDVAL network of sites (Section 5.2). However, some spatial and temporal inconsistencies were found around the globe. In particular, the temporal analysis showed unrealistic drops, unrealistic peaks in winter and late ending of the growing season. These issues mainly affect to forest classes in which higher discrepancies are observed with regard to the considered satellite products. The accuracy assessment over a limited number of sites (N=28) showed an overall accuracy (RMSD) of 0.13 for FVC products, 0.56 for LAI and 0.13 for FAPAR with slight mean positive bias for all variables. EPS VEGA showed good accuracy for LAI (RMSD=0.56), similar to MODIS LAI Collection 6. For FVC, an accuracy of RMSD=0.13, providing slightly better results than PROBA-V V1. For FAPAR, an accuracy of RMSD=0.13 was found, similar to PROBA-V V1 but worse than MODIS FAPAR Collection 6. The extended data record (2007–2021) including Metop-A, demonstrated good stability comparable to MODIS, with similar accuracy RMSD of 0.77 (LAI) and 0.12 (FAPAR/FCOVER) against GBOV V2 (N>400) [VR-EPSVEGA-R].

- NASA MODIS (MxD15) LAI/FAPAR Collection 6.1

Terra and Aqua MODIS LAI/FAPAR (MxD15) Collection 6 (C6.1), available since July 2002 from https://lpdaac.usgs.gov/products/ is produced at 500 m spatial resolution and 8 days step over a global sinusoidal grid (Yan et al., 2016a). The algorithm chooses the "best" pixel available from all

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the acquisitions of both MODIS sensors located on NASA's Terra satellites within the 8-day period. The main algorithm is based on Look Up Tables (LUTs) simulated from a three-dimensional radiative transfer model (Knyazhikin et al., 1998). The MODIS red and NIR atmospherically corrected reflectances (Vermote et al., 1997) and the corresponding illumination-view geometry are used as input for the LUTs. The output is the mean LAI/FAPAR computed over the set of acceptable LUT elements for which simulated and measured MODIS surface reflectances are within specified uncertainties.

The accuracy assessment of MxD15 C6, performed over 45 FAPAR ground measurements showed an overestimation over sparsely vegetated areas (Yan et al., 2016b). Comparisons with SPOT/VGT 1km V1 products showed similar spatial distributions at a global scale (Yan et al., 2016b), and temporal comparisons for the 2001–2004 period showed that the products properly captured the seasonality of different biomes, except in evergreen broadleaf forests. Fuster et al. (2020) reported an accuracy (RMSD) of 0.91 for LAI and 0.14 for FAPAR, performed against ground measurements over a four-year period (2014–2017), mainly over crops.

- NASA VIIRS (VNP15) LAI/FAPAR Version 2.

As Terra and Aqua MODIS sensors will likely be terminated (Terra and Aqua MODIS have far exceeded their design life, 6 years, and have a strong chance of operating successfully into the early 2020s.), the Visible/Infrared Imager Radiometer Suite (VIIRS) instrument onboard the Suomi National Polar-orbiting Partnership (SNPP) and Joint Polar Satellite System (JPSS) has inherited the scientific roles of MODIS (Justice et al., 2013). The first VIIRS sensor onboard the SNPP platform was successfully launched in October 2011. In this context, the NASA SNPP VIIRS LAI/FAPAR product (VNP15) (https://viirsland.gsfc.nasa.gov/Products/NASA/LAI_FparESDR.html) should ensure the continuity with the MODIS LAI/FAPAR product (MxD15).

The VIIRS LAI/FPAR algorithm [ATBD-VIIRS-LAI_FPAR] has benefitted from the heritage of the MODIS operational algorithm. The algorithm retrieves LAI and FPAR values given sun and view directions, Bidirectional Reflectance Factor (BRF) for each spectral band, uncertainties (i.e., relative stabilized precision, Wang et al., 2001) in input BRFs, and land cover classes based on a 8-biome classification map (Myneni et al., 2002; Yang et al., 2006). The operational VIIRS LAI/FPAR algorithm consists of a main algorithm that is based on 3D radiative transfer equation. By describing the photon transfer process, this algorithm links surface spectral BRFs to both structural and spectral parameters of the vegetation canopy and soil (Asrar and Myneni, 1991; Ross, 2012). Given atmosphere corrected BRFs and their uncertainties, the algorithm finds candidates of LAI and FAPAR by comparing observed and modeled BRFs that are stored in biome type specific LUTs. All canopy/soil patterns for which observed and modeled BRFs differ within biome-specified thresholds

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of uncertainties (e.g., 30% and 15% for red and near-infrared bands, respectively, for forest biomes) are considered candidate solutions and the mean values of LAI and FAPAR from these solutions are reported as outputs. The mean and dispersion of LAI/FAPAR candidates are reported as retrieval and its reliability, respectively. The law of energy conservation and the theory of spectral invariance are two important features of this main algorithm [ATBD-MOD15].

The validation approach for VIIRS LAI/FAPAR is similar to that of MODIS Collection 6. The VIIRS subsets and the EOS Core Validation Sites as well as the BELMANIP2 sites are used to achieve the goal of CEOS level 2 validation stage. LAI and FAPAR retrievals from VIIRS and MODIS were found consistent at different spatial (i.e., global and site) and temporal (i.e., 8-day, seasonal and annual) scales (Xu et al., 2018), with mean discrepancies (-0.006 ± 0.013 for LAI and -0.002 ± 0.002 for FPAR) meeting the stability requirement for long-term LAI/FAPAR Earth System Data Records from multi-sensors as suggested by the GCOS. Xu et al., (2018) also reported uncertainties (relative uncertainties) of VIIRS LAI and FAPAR products, assessed through comparisons to ground measurements, of 0.60 (42.2%) and 0.10 (24.4%), respectively. Comparison of VIIRS LAI/FAPAR (VNP15A2H) versus GBOV ground data over North America sites (Brown et al., 2020) showed similar performance than MODIS Collection 6 (MOD15A2H), with RMSD = 0.81 to 0.89 for LAI, and RMSD = 0.12 for FAPAR. More recently (Román et al., 2024) MODIS C6.1 and VIIRS V2 LAI/FAPAR products showed very high consistency ($R^2 > 0.95$) with no systematic bias, supporting strong continuity.

- C3S PROBA-V V3 and Sentinel-3 LAI/FAPAR V4.1 (C3S PBV V3 + S3 V4.1)

Copernicus Climate Change Service (C3S) provides different versions of LAI and FAPAR products depending on the processing algorithm and/or input datasets.

- C3S PROBA-V LAI/FAPAR V3 (C3S PBV V3)

C3S LAI and FAPAR PROBA-V based multi-sensor V3 products are generated at 1 km of spatial resolution using the Two-stream Inversion Package (TIP) algorithm (Pinty et al., 2006), which provides effective LAI and FAPAR along with their uncertainties by applying the TIP to VIS and NIR broadband albedos. More details on the retrieval procedure can be found in the ATBD [ATBD-C3S_V3]. The input data is C3S multi-sensor surface albedo version 2.0 (Carrer et al., 2021), which incorporates harmonized measurements from the platforms National Oceanic and Atmospheric Administration (NOAA-[7…17]), SPOT/VGT and PROBA-V, which result in almost 40 years of satellite observations. The algorithm uses a composite period of 20 days using recursive temporal composition scheme (Carrer et al., 2021). A Z_Age field is provided as ancillary layer, representing the "age" of the information or elapsed time (in days) of the last available observation used for each image pixel. Multi-sensor surface albedo harmonization is achieved at the BRDF retrieval step, using

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a Kalman filter and BRDF model with a SPOT/VGT-based BRDF climatology to reduce gaps and improve cross-sensor consistency.

The validation report of C3S LAI/FAPAR V3 [PQAR-C3S_V3] showed good completeness and reliable values. A systematic negative bias in C3S FAPAR V3 values, ranging between -30% and -20%, was found when compared to satellite-based references (CLMS PBV 1km V2, NASA VIIRS and LSA SAF AVHRR). The comparison with ground data showed overall good results for FAPAR (but underestimation of highest values) and systematic negative bias for C3S effective LAI V3 products. Another limitation is the high density of points with LAI and FAPAR very close to 0 (mainly in winter at northern latitudes).

- C3S Sentinel-3 LAI/FAPAR V4.1 (C3S S3 V4.1)

C3S S3 LAI and FAPAR V4.1 products are generated at 300 m resolution using the same TIP algorithm (Pinty et al., 2006) [ATBD-C3S_V4.1] as C3S PBV V3. The input data is C3S S3 surface albedo V3.1 (Sánchez-Zapero et al., 2023b), based on CLMS TOC reflectance V2.3 (i.e., as in CLMS OLCI LAI/FAPAR/FCOVER V2) but using both OLCI and SLSTR Sentinel-3A and -3B data. The C3S surface albedo V3.1 algorithm is based on BRDF inversion from the Regularised BRDF inversion for Land Surface (ReBeLS) processor (León-Tavares et al., 2024), which uses prior MODIS BRDF climatology. The BRDF descriptor from ReBeLS are angularly integrated to generate spectral albedo quantities for the selected OLCI (Oa03, Oa04, Oa07, Oa17 and Oa21) and SLSTR (S1, S2, S5 and S6) bands. Finally, the spectral integration module infers broadband albedo quantities for three standard broad bands (visible, NIR and total shortwave).

C3S LAI/FAPAR V4.1 shows good temporal consistency and continuity with the predecessor C3S PBV V3, but with a systematic positive bias (~10%) due to the change of input data (PROBA-V to S3) and product version [PQAR-C3S_V4.1]. A key limitation is the high frequency of near-zero values in winter at latitudes >45ºN like for C3S PBV V3. Validation against quality-controlled GBOV FAPAR land products showed a slight negative mean bias of -0.07 with RMSD of 0.11. LAI showed underestimation compared with effective LAI values over homogeneous forest sites (B=-0.99).

Since C3S LAI corresponds to an effective LAI, it is only used in this exercise to qualitatively assess seasonal and interannual variation, whereas C3S FAPAR is used for both qualitative and quantitative analyses.

➢ JRC ESA Sentinel-3/OLCI GIFAPAR

The OLCI level-2 land products include the Green Instantaneous FAPAR products (GIFAPAR), updated name of former OLCI Global Vegetation Index (OGVI). GIFAPAR is the continuity of the MERIS FAPAR (MERIS Global Vegetation Index (MGVI)) [ATBD-MGVI]. The GIFAPAR

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algorithm [ATBD-GIFAPAR] consists of two steps. First, the bidirectional reflectance factors are "rectified", that is, angular effects are removed. Then, the information from the blue channel is used to decontaminate the red and near-infrared bands from any atmospheric influence. Such an approach does not require any assumption on the ambient atmospheric properties (Gobron et al., 2015). Main difference of the OLCI products compared with MODIS and MERIS is the data composition algorithm. OGVI used the actual value for the day while MOD15 and MGVI were composited using the arithmetic mean and maximum value across eight days, respectively (Gobron et al. 2015).

Seasonality comparisons against ground-based measurements have shown that OLCI FAPAR effectively describes the seasonal changes over various types of crops, and more ground-truth is under collection for validation purposes (Zhang et al., 2020).

4.2.3 GPP

The Numerical Terradynamic Simulation Group (NTSG) of the University of Montana provides GPP and NPP from MODIS [MOD17-UG] (http://www.ntsg.umt.edu/project/mod17). The MOD17A2HGF Version 6.1 dataset is a GPP product at 500m resolution. It is a gap-filled 8-daily product based on 8-daily composites of FAPAR and LAI derived from the MODIS MOD15 LAI/FAPAR data product at 500m. The MOD17 algorithm [ATBD-MOD17] is a satellite-based Production Efficiency Model (PEM) of the form of Monteith (1972), which suggests that productivity of annual crops under well-watered and fertilized conditions is linearly related to the amount of Absorbed Photosynthetically Active Radiation (APAR) (Zhao et al. (2005). The translation of APAR to an actual productivity estimate (GPP) is conducted via a conversion efficiency parameter, ε, which varies by vegetation type and climate conditions. Temperature, incoming solar radiation, and vapour pressure deficit are derived from a meteorology dataset. Meteorology datasets used by various versions of the MOD17 algorithm include products from the NASA Global Modelling and Assimilation Office and the NCEP/NCAR Reanalysis II. The MODIS MCD12Q1 data product is used as a land cover classification.

The MODIS GPP has been validated against FLUXNET measurements (Reichstein et al., 2002) and BigFoot estimates (Turner et al., 2006). In the first validation, monthly GPP values of the 101*101 pixels (1km) surrounding the flux tower were averaged and compared with the FLUXNET values. The largest bias occurs in the higher GPP values with an underestimation for MODIS product. Possible causes for the bias can be attributed to a poor model performance for the vapor pressure deficit response, soil drought effects, high radiation conditions, and an underestimation of the GPP in the most productive stands. Reichstein et al. (2002) also gives the RMSE for the comparison of the MODIS data with FLUXNET values for six individual towers: on an 8-daily time step the RMSE goes from 34 kg DM/ha/day for the best performing site up till 63 kg DM/ha/day for the site with the

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highest error. The comparisons of annual MODIS GPP with BigFoot data show close agreement at 2 boreal and hardwood forests sites, overestimation at 2 conifer and mixed forest sites, and large underestimation at one tropical moist forest site (Turner et al., 2006).

The MODIS MOD17 V6.1 GPP datasets used in the validation are global sub-sampled images, a selection of 10° x 10° tiles at full resolution, and extracts over the LANDVAL V2 sites (Section 5.2) over the years 2019 and 2023 for statistical and spatial analyses, and over the period 2014 – 2024 for temporal analysis.

4.2.4 NDVI

- 300m NDVI V1

The 300m NDVI V1 is a standard 10-day directional NDVI synthesis product, based on maximum value compositing over a 10-day period. The time series of NDVI V1 is derived from PROBA-V Collection 1 (Toté et al., 2018) 10-daily Top-of-Canopy (TOC) composites (January 2014 – June 2020).

The NDVI 300m V1 datasets used in the validation are global sub-sampled images, a selection of 1° x 1° and 10° x 10° tiles at full resolution, and extracts over the LANDVAL V2 sites (Section 5.2) for the period 2019 for statistical and spatial analyses, and over the whole time series (January 2014 – June 2020) for temporal analysis.

- 300m NDVI V2

The NDVI 300m V2 products are BRDF-normalized NDVI products derived from Sentinel-3/OLCI TOC reflectance [CGLOPS1_ATBD_S3-AC-V1.1, CGLOPS1_ATBD_BRDFCorrection300m-V1.6]. The time series of NDVI 300m V2 spans from July 2020 to the present. The dataset is described in more detail in [CGLOPS1_ATBD_NDVI300m-V2].

The Sentinel-3/OLCI based NDVI 300m V2 datasets used in the validation are global sub-sampled images, a selection of 1° x 1° and 10° x 10° tiles at full resolution, and the LANDVAL V2 sites (Section 5.2) over the year 2023 for statistical and spatial analyses, and over the period July 2020 – December 2024 for the temporal analysis.

- 1km NDVI V3

The NDVI 1km V3 products are BRDF-normalized NDVI products generated from the SPOT/VEGETATION Collection 3 (Toté et al., 2017) and PROBA-V Collection 1 (Toté et al., 2018) data. The time series include data from different sensors, i.e. SPOT4/VGT1 (May 1998-January 2003), SPOT5/VGT2 (February 2003 – December 2013), and PROBA-V (January 2014 – June 2020).

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The SPOT4 and SPOT5 satellites have a local equator-crossing time of around 10:30 AM. The overpass time for PROBA-V at launch was 10:45 AM. Since PROBA-V has no on-board propulsion, natural drift has led to consequent earlier overpass times, till around 9:20 AM by the end of the operational mission in June 2020. A spectral harmonisation is applied on the NDVI derived from SPOT4/VGT1. This is explained in more detail in [CGLOPS1_ATBD_NDVI1km-V3].

The results of the exhaustive scientific validation of NDVI 1km V3 are described in [CGLOPS1_QAR_NDVI1km-V3].

The NDVI 1km datasets used in the validation are global sub-sampled images and extracts over the LANDVAL V2 sites for the period 2019 for statistical and spatial analysis, and for the period 2014 – 2024 for temporal analyses.

- MODIS MCD43A4 NDVI

The MODerate Resolution Imaging Spectroradiometer (MODIS) is mounted on two platforms, EOS-TERRA and AQUA. The MODIS instruments on Terra and Aqua capture data in 36 bands with a spatial resolution of 250m (2 bands), 500m (5 bands) and 1 km (29 bands). Both instruments image the same area on Earth approximately three hours apart: Terra has a morning equator overpass (descending node) at around 10:30 local time, while Aqua is timed to cross the equator (ascending node) at approximately 13:30 local time.

The MCD43A4 Version 6.1 dataset is a nadir BRDF-adjusted reflectance at 500m resolution, adjusted to the solar zenith angle at local solar noon (Schaaf et al., 2011, 2002). The MCD43A4 reflectance represents the best characterization of the surface possible from the inputs available over a 16-day period from both Terra and Aqua data. The NDVI is calculated by combining the reflectances of band 1 (red) and 2 (NIR).

MCD43A4 is provided in Sinusoidal projection, hence the MODIS MCD43A4 NDVI was resampled to the CLMS products 1km grid in geographical projection system, based on a cubic resampling method. Since the MCD43A4 dataset is produced every day (with 16 days of acquisition, i.e. product date +/- 8 days), the temporal resolution was adapted to the temporal resolution of the CGLOPS NDVI products (10-daily) by considering three products per month, each with maximum overlap with the compositing window of the CGLOPS NDVI products. Quality flags (missing/invalid, snow, water) were derived from the related MCD43A2 dataset describing the overall condition of the BRDF product. Data based on less than 2 observations for BRDF inversion, data over water, and snow pixels were excluded from the analysis.

The MODIS NDVI datasets used in this study are global sub-sampled images, a selection of 10° x 10° tiles at full resolution (i.e. 1km resample), and extracts over the LANDVAL V2 sites for the

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period 2019 and 2023 for statistical and spatial analyses. The period 2014 – 2024 is used for temporal analyses.

4.2.5 Land Surface Temperature

In the case of LST, comparisons with independent satellite-based LST products are useful, not as validation *per se* (since none of the satellite products represents the absolute "truth"), but as an approach to assess the spatial and temporal consistency of the evaluated dataset. For the CLMS LST product, this analysis is particularly relevant due to the integration of multiple geostationary missions covering different regions of the globe, allowing the assessment of potential spatial differences between the corresponding geostationary domains.

In addition to cross-comparisons with previous CLMS LST product versions, the assessment of CLMS LST V3.0 is further carried out using LSTs derived from the Sea and Land Surface Temperature Radiometer (SLSTR) onboard the ESA Sentinel-3A satellite. The Sentinel-3 SLSTR LST product provides independent polar-orbiting observations at high spatial resolution and is therefore suitable for evaluating the spatial consistency of the geostationary-based CLMS LST dataset. The quality of the SLSTR LST product is documented in the corresponding VR [S3MPC.UOL.VR.029]. Previous validation exercises indicate good agreement with ground-based observations and consistency with other satellite LST products, with differences generally within ~1 K, except over areas affected by surface heterogeneity, large viewing angles, and challenging observation conditions.

4.2.6 Burned Areas

The intercomparison analysis includes three external global Burned Areas products:

1. The MODIS-MCD64 Collection 6 product (Giglio et al., 2018): MCD64 is a NTC product derived from 500m MODIS data and active fire detections from the Collection 6 Terra MOD14A1 and Aqua MYD14A1 products. The product algorithm first generates images of persistent changes of a burn-sensitive vegetation index, which are then classified as burned / unburned with probabilistic thresholds drawn from samples at active fire detections
2. the ESA's Climate Change Initiative Fire Disturbance project FIRECCIS311 product (https://climate.esa.int/en/catalogue/d441079fc77f49fabeb41330612b252f/) is an NTC product derived from 300m Sentinel-3 SLSTR data and active fire detections from the NASA's VIIRS Active Fire and Thermal Anomalies product. The retrieval methodology uses a contextual growing algorithm based on shortwave infrared reflectance thresholds

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dynamically selected on each processing 10ºx10º tile (Lizundia-Loiola et al., 2022). FIRECCIS311's algorithm included several improvements respect to C3S burned areas detection approach, mainly on the temporal compositing criteria, on the definition of dynamic local thresholds and on the use of a spectral index based on two SWIR spectral bands instead of a single NIR reflectance band.

4.2.7 Land Surface Phenology

- Global products.

For product intercomparison with LSP SOSD and EOSD layers, two different global products are used:

1. NASA MODIS Land Cover Dynamics Version 6.1 (MOD12Q2 V6.1)

The Terra and Aqua combined Moderate Resolution Imaging Spectroradiometer (MODIS) Land Cover Dynamics (MCD12Q2) Version 6.1 (https://lpdaac.usgs.gov/products/mcd12q2v061/) data product provides global land surface phenology metrics at yearly intervals from 2001 (Zhang et al., 2006) [MOD12Q2_User_Guide_V61]. The MCD12Q2 Version 6.1 data product is derived from time series of the 2-band Enhanced Vegetation Index (EVI2) calculated from MODIS Nadir BRDF-NBAR. Vegetation phenology metrics at 500-meter spatial resolution are identified for up to two detected growing cycles per year. For pixels with more than two valid vegetation cycles, the data represent the two cycles with the largest NBAR-EVI2 amplitudes. For areas where the NBAR-EVI2 values are missing due to cloud cover or other reasons, the data gaps are filled with good quality NBAR-EVI2 values from the year directly preceding or following the product year.

The layers provided by the product correspond to the total number of vegetation cycles detected for the product year (up to two cycles/year), the onset of greenness, greenup midpoint, maturity, peak greenness, senescence, greendown midpoint, dormancy, EVI2 minimum, EVI2 amplitude, integrated EVI2 over a vegetation cycle, as well as overall and phenology metric-specific quality information. MCD22Q2 V6.1 detects and provides different phenological dates using 15% (Greenup and Dormancy for start and end of season), 50% (MidGreenup and MidGreendown), and 90% (Maturity and Senescence) EVI2 amplitude.

Comparison of vegetation phenology retrieved from MODIS with in-situ measurements revealed realistic estimates over Harvard Forest (Zhang et al., 2006). More generally, the spatial distribution of phenological metrics estimated from MODIS data is qualitatively realistic, and exhibits strong correspondence with temperature patterns in mid- and high-latitude climates, with rainfall seasonality in seasonally dry climates, and with cropping patterns in agricultural areas (Zhang et al., 2006).

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2. NASA VIIRS Land Cover Dynamics Version 2 (VNP22Q2 V2)

The Suomi National Polar-Orbiting Partnership (Suomi NPP) NASA Visible Infrared Imaging Radiometer Suite (VIIRS) Land Cover Dynamics (https://lpdaac.usgs.gov/products/vnp22q2v001/) data product provides global land surface phenology metrics at yearly intervals (Zhang et al., 2018b). The VNP22Q2 V2 data product is derived from time series of EVI2 calculated from VIIRS Nadir BRDF-NBAR. Vegetation phenology metrics at 500-meter spatial resolution are identified for up to two detected growing cycles per year. VIIRS LCD V2 products are generated with same algorithm as previous V1 [VNP22_User_Guide_V1], but introduces some improvements or changes:

- Improved calibration algorithm and better coefficients for entire SNPP mission.
- Improved geolocation accuracy and updates to fix outliers around maneuver periods and other events.
- Corrections to the aerosol quantity flag (low, average, high) mainly over brighter surfaces in the mid to high latitudes such as desert and tropical vegetation areas. This has an impact on the retrieval of other downstream data products such as VIIRS Vegetation Indices and BRDF/Albedo.
- Improved cloud mask input product for corrections along coastlines and artifacts from use of coarse resolution climatology data.
- Replaced the land/water mask input product with MODIS heritage seven class land/water mask.

The main changes respect to the predecessor (MCD12Q2 V6.1) are first, VNP22Q2 reconstructs the EVI2 time series using a physically-based hybrid piecewise logistic model (HPLM), while MCD12Q2 V6.1 fits the EVI2 time series using a penalized cubic smoothing spline. Second, VNP22Q2 identifies phenological transition dates using the curvature change rate derived from the HPLM without any predefined thresholds involved, whereas MCD22Q2 V6.1 detects phenological dates using thresholds of EVI2 amplitude. Third, VNP22Q2 records the phenological metrics occurring within a given calendar year or product year however, MCD12Q2 V6.1 records the metrics for the vegetation cycles for which the peak date falls within the calendar year of the product year.

The product contains six phenological transition dates: onset of greenness increase, onset of greenness maximum, onset of greenness decrease, onset of greenness minimum, dates of mid-greenup, and mid-senescence phases. The product also includes the growing season length. The greenness related metrics consist of EVI2 onset of greenness increase, EVI2 onset of greenness maximum, EVI2 growing season, rate of greenness increase and rate of greenness decrease. The confidence of phenology detection is provided as greenness agreement growing season, proportion of good quality (PGQ) growing season, PGQ onset greenness increase, PGQ onset greenness maximum, PGQ onset greenness decrease, and PGQ onset greenness minimum. The final layer is

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quality control specifying the overall quality of the product. A low-resolution browse image showing greenup is also available when viewing each VNP22Q2 granule.

Comparison of VNP22Q2 resulting phenological metrics over United States with species-specific field phenological observations, Landsat phenology retrievals and the MODIS phenology detections showed that the VIIRS phenometrics ar of high quality and are in a good agreement with the other independent satellite and field observations (Zhang et al., 2018b). Globally, the phenological dates derived from VIIRS EVI2 were compared with MODIS detections (Zhang et al., 2018a), which showed a good agreement with an average absolute difference less than a week except for the senescence onset. Phenological dates derived from VIIRS EVI2 and PhenoCam were generally comparable for the vegetation greenup phase, but differed considerably for the senescence phase (Zhang et al., 2018a). The continuity between VIIRS V2 and MODIS V6.1 phenometrics was assessed and they agreed well and had little systematic bias (Román et al., 2024).

- Regional product.

Additionally, the High-Resolution Vegetation Phenology and Productivity product suite (HR-VPP) produced within the Pan-European component of the Copernicus Land Monitoring Service (CLMS) is used for regional comparisons over Europe:

HR-VPP products are derived from the Sentinel-2 constellation at high spatial resolution (10m x 10m with a revisit time of 5 days). They are generated over the entire EEA39 (32-member countries, the UK and 6 cooperating countries in the Western Balkans) from January 1, 2017 onwards.

The HR-VPP product suite contains 3 product groups, 31 product types, 1522 files and more than 900.000 tiles per year, which totals more than 80 Terra Bytes of data, per year. The three main groups of products, being (in order of production), are:

- The **raw Vegetation Indices** (VIs). This group consists of products generated near real-time (NRT) providing for every pixel the status of the vegetation vigor. The status is provided in four ways: LAI, FAPAR, NDVI and PPI;
- The **Seasonal Trajectories** (STs) products are provided yearly after the end of the vegetation growing season. These are derived as a regular time-series of every 10 days by fitting a smoothing and gap filling function to the raw Plant Phenology Index, generated in the product group Vis. TIMESAT version 4 software package (Jönsson and Eklundh, 2004) is used to produce STs from PPI retrievals;
- The **Vegetation Phenology Parameters** (VPPs) products bundle is derived from the STs of the PPI index, on a yearly basis, after the end of the growing season. VPP metrics are provided for up to two growing seasons, being e.g. start of the season, end of season,

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seasonal productivity, etc. These metrics are defined in concordance with CLMS LSP products, following similar algorithm (Tian et al., 2021).

Validation of HR-VPP phenology metrics for 2021-2024 period [CLMS_HRVPP_VR_ST-VPP] showed good agreement with existing medium-resolution products (MODIS LCD V6.1 and VIIRS LCD V1) over Europe. The accuracy of phenology metrics (SOSD, EOSD) as compared with ground PEP725 observations showed a slight positive bias of 4 days for SOSD (i.e., late start of season) and negative bias of -6 days for EOSD (i.e., early end of season).

4.2.8 Evapotranspiration

The ETA and HF products were compared with other satellite-based products which contain, at least, one of the variables produced within CLMS. The products used for intercomparison were:

- WaPOR ETA (https://wapor.apps.fao.org/): The data layers of interest for this analysis are the actual evapotranspiration, the transpiration and evaporation, in particular the Version 3 Level 1 products which exhibit the same spatial and temporal features as the CLMS ETA product.
- The MOD16A2 product (https://www.earthdata.nasa.gov/data/catalog/lpcloud-mod16a2gf-061). This product is based on observations by the MODIS sensor onboard the Terra and Aqua satellites and reanalysis meteorological data. The data are available at 500 m spatial resolution and a time step of 8-days; two separate datasets are distributed corresponding to the two satellites carrying the MODIS sensor.
- LSA-SAF ET/H/LE (https://lsa-saf.eumetsat.int/en/data/products/evapotranspiration-and-turbulent-fluxes/): This product is based on observations by the EUMETSAT's Meteosat Second Generation satellite. This product covers Europe, Africa and Eastern South America. Although this product does not have a global coverage, it is particularly interesting as it contains half-hourly estimates of latent and sensible heat fluxes. Therefore, it allows intercomparison with the CLMS HF products which are produced at satellite overpass time.
- GLEAM 4.2 (https://www.gleam.eu/): GLEAM stands for “Global Land Evaporation Amsterdam Model” and is a global product delivering daily estimates of land evaporation and separate components (transpiration, interception loss, snow sublimation) and root-zone soil moisture. The estimation of land evapotranspiration departs from the estimation of potential evapotranspiration on the basis of the Penman method and then, accounting for stress factors related to soil moisture and vegetation traits. The product is available at 0.1 spatial resolution at a daily temporal step.

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4.3 MODELLED REFERENCE PRODUCTS

4.3.1 Global Land Data Assimilation System

Modelled reference products to which the CLMS soil moisture products are compared were taken from the GLDAS (Global Land Data Assimilation System) Noah-LSM (Noah Land Surface Model), version 2.1. The GLDAS Noah Land Surface Model (Rodell et al., 2004, Beaudoing et al., 2020) is produced by NASA. It includes four soil layers with the bottom depth at 0.1, 0.4, 1.0 and 2.0 m respectively, and simulates over 30 land surface parameters in total. Data is available with a temporal sampling of three hours on a global 0.25° grid. The model soil moisture data was converted from kg/m^2 to m^3/m^3 by using the formula $SM[m^3/m^3] = SM[kg/m^2] * 0.001 * 1/d$, where d is the thickness of the soil layer in meters. The factor 0.001 is due to the assumption that 1kg of water represents $1000cm^3$, which is $0.001m^3$.

The GLDAS model dataset is chosen for the validation of the CLMS soil moisture products because it provides consistent long term, global data and is updated regularly. The consistency of the dataset also makes it likely that future quality evaluation exercises can use them as well. This would have been less likely if we had chosen e. g. soil moisture products derived from other remote sensing platforms that are vulnerable to satellite failure etc. Other reanalysis products with comparable consistency, record length, and spatial coverage may serve equally as modelled reference.

Modelled reference data is of value for the validation of the kilometric scale products. Temporal soil moisture dynamics are largely driven by atmospheric forcing, foremost precipitation and evapotranspiration, which acts at scales much coarser than one kilometre (Vachaud et al., 1985 and Wagner et al., 2008). Reanalysis models ingest this forcing directly and provide temporally complete and consistent records, with errors that are independent of the retrieval errors of satellite sensors. Consequently, coarse scale model data provides a robust reference for the temporal signal of the 1km products, for statistics aggregated over many locations.

4.3.2 ERA5 reanalysis

The temporal stability of the CLMS LST V3.0 product is assessed using the ERA5 (Hersbach et al., 2020) skin temperature reanalysis product. ERA5 skin temperature datasets are produced by the ECMWF and are obtained from the C3S Climate Data Store (https://cds.climate.copernicus.eu/). This surface parameter has been shown to provide a suitable large-scale proxy for satellite-derived LST consistency assessments (Trigo et al., 2015).

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5 PRODUCT VALIDATION METHODOLOGIES

This chapter explains the approaches used to validate the CLMS (group of) products. It describes the criteria of evaluation and how the reference datasets presented in the previous chapter are used to quantify these criteria.

5.1 TOC REFLECTANCES

The general premise behind the current state-of-art surface reflectance validation approach is to compare the surface reflectance derived from Atmospheric Correction (AC) using the image derived Aerosol Optical Thickness against the equivalent using the station derived AOT. This assumes that the surface reflectance derived from the station AOT is the "truth" (Vermote et al., 2016). This is the approach performed on the quality assessment of the AC of the TOA to TOC pre-processing chain [CGLOPS1_QAR_S3-AC-V1.1]. However, some common themes begin to emerge when analyzing the studies using these networks. Firstly, the combined impact of the sensor spatial resolution (Sentinel-3, 300m in this case) and the site heterogeneity can cause significant disagreement. Secondly, there is little mention or quantitative assessment of the uncertainties associated with the in-situ data nor any endeavour to perform the validation in a metrologically robust way. Thirdly, many of the published activities affecting many sensors (for example: Sentinel-2: Gascon et al., 2017; MODIS: Vermote et al., 2016; and AATSR: Sogacheva et al., 2015) utilize reference data which cannot be considered independent (e. g. those utilizing other satellite products). This is because the main assumption is that the aerosol correction is the dominant error. This may be the case, but by only comparing the impact of the aerosol retrieval, the impact of other errors in the processing chain (radiometric, geolocation, etc.) is lost and the validation is only testing the aerosol retrieval. For that reason, validation satellite surface reflectance data is conducted against independent in-situ reflectance estimates (e. g. Li et al., 2010; Liang et al., 2002). The CEOS RadCalNet and ESA FRM are used as independent in-situ data (Section 4.1.1).

Validation of surface reflectance should be taken in the context of the validation of the related products that are the albedo and the BRDF (Bidirectional Reflectance Distribution Function). All three fall within the same category in the CEOS LPV. For that, the validation methodology follows, as much as possible, the CEOS LPV good practice protocol for the validation of satellite-derived albedo products (Wang et al., 2019). Two main validation approaches are defined:

- The direct point-to-pixel validation, which involves comparisons of satellite products with TOC reflectance measured from in-situ tower-based instruments (i. e. direct validation).
- Product-to-product validation approach (e.g., Sentinel-3 versus Sentinel-2), referred to satellite product intercomparison or indirect validation. The intercomparison offers a means of assessing discrepancies between products and provides relative uncertainties. It is very helpful

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due to limited number of ground stations, and Areas Of Interest (AOI) are selected over different biomes and latitudes (e.g., snow, desert, tropical forest, agriculture areas).

The following criteria are analysed: completeness, spatial consistency, temporal consistency and Accuracy, Precision and Uncertainty (APU).

(1) **Completeness** corresponds to the absence of spatial and temporal gaps in the data. Missing data are mainly due to cloud or snow contamination, poor atmospheric conditions or technical problems during the acquisition of the images, and is generally considered by users as a severe limitation of a given product. It is therefore mandatory to document the completeness of the product (i.e. the distribution in space and time of missing data).
   - Monthly composites global maps of missing values are computed.
   - Distribution of gaps as a function of the season.

(2) **Spatial consistency** refers to the realism and repeatability of the spatial distribution of retrievals over the globe.
   - The methodology includes the visualization of animations of global maps at a reduced resolution and visualization of zooms over selected AOIs at full resolution. Qualitative check of the realism and repeatability of spatial distribution of retrievals and the absence of artifacts (tiles, stripes, outliers, etc) based on the expert knowledge of the scientist is performed. The ancillary layers (quality flags and uncertainties) are also displayed.
   - Difference maps versus references over selected AOIs are computed in order to evaluate the spatial distribution of the differences.

(3) **Temporal consistency** refers to the realism and stability of the inter-annual and intra-annual temporal variations.
   - Qualitative inspection of the temporal variations of the products is evaluated over selected AOIs and sites with ground information (e.g., RadCalNet, FRM (Section 4.1.1)).
   - To quantitative analyze the temporal consistency of the products, the cross-correlation of the temporal series is assessed. Cross-correlation is a standard method of estimating the degree to which two series are correlated. Considering two series x(i) and y(i) where i=0,1,2...N-1, the cross correlation ρ at delay d is defined as:

$$\boldsymbol{\rho} = \frac{\sum_i[(\mathbf{x(i)}-\mathbf{mx})\cdot(\mathbf{y(i}-\boldsymbol{d})-\mathbf{my})]}{\sqrt{\sum_i(\mathbf{x(i)}-\mathbf{mx})^2}\sqrt{\sum_i(\mathbf{y(i}-\boldsymbol{d})-\mathbf{my})^2}} \qquad \textbf{Eq. 1}$$

   where mx and my are the means of the corresponding series. The cross-correlation is computed with a delay d=0 in order to estimate the consistency between temporal profiles of different products with the same temporal support.

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(4) **Accuracy**, **Precision** and **Uncertainty** (**APU**) are quantified by several metrics reporting the goodness of fit between the products and the corresponding reference dataset. Accuracy is quantified by the bias and Median Deviation (MD - according to the LPV best practice). Precision represents the dispersion of product retrievals around their expected value and can be estimated by the Standard Deviation (STD) of the difference between retrieved satellite product and the corresponding reference estimates. Median Absolute Deviation (MAD) is also calculated to report to quantify the precision, according to the CEOS LPV best practices. Uncertainty includes systematic and random errors and can be estimated by the Root Mean Square Deviation (RMSD). In addition to these metrics, other statistics are useful to evaluate the goodness of fit between two datasets including linear model fits. For this purpose, Major Axis Regression (MAR) is computed instead of ordinary least square regression because is specifically formulated to handle error in both of the x and y variables (Harper, 2014). These metrics are summarized in Table 2.

- Scatter-plots and validation metrics versus references are produced.

Table 2: Validation metrics

Statistics	Comment
N	Number of samples. Indicative of the power of the validation
B	Mean Bias. Difference between average values of x and y. Indicative of accuracy and offset. Bias (%) is the relative mean bias between the average of x and y.
MD	Median deviation between x and y. CEOS LPV best practice reporting the accuracy. MD (%) is the relative MD between the average of x and y.
STD	Standard deviation of the pair differences. Indicates precision.
MAD	Median absolute deviation between x and y. CEOS LPV best practice reporting the precision. MAD (%) is the relative MAD between the average of x and y.
RMSD	Root Mean Square Deviation. RMSD is the square root of the average of squared errors between x and y (see Table 6 for formulae). CEOS LPV best practice reporting uncertainty. RMSD (%) is the relative RMSD between the average of x and y.
R	Correlation coefficient. Indicates descriptive power of the linear accuracy test. Pearson coefficient is used.
MAR	Slope and offset of the Major Axis Regression linear fit. Indicates some possible bias
p-value	Test on whether the slope is significantly different to 1.

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Table 3 summarizes the validation criteria evaluated for TOC Reflectances, and the methods and metrics used to assess each criterion.

Table 3: Summary of criteria and validation metrics of TOC Reflectances

Criterion	Methods and validation metrics
Completeness	Monthly global maps of missing values. Distribution of gaps as a function of the season.
Spatial Consistency	Visual inspection of global maps and selected AOIs and check of ancillary layers. Difference maps over selected AOIs.
Temporal Consistency	Qualitative inspection of temporal variations. Cross-correlation.
Accuracy	Scatter-plots vs references (N, Bias, MD, R, MAR, p-value).
Precision	Scatter-plots (STD, MAD).
Uncertainty	Scatter-plots (RMSD).

5.2 LAI, FAPAR, FCOVER

The validation methodology follows standardized procedures for satellite-derived biogeophysical products (Camacho et al., 2025) fully compliant with the CEOS LPV LAI validation good practices protocol (Fernandes et al., 2014) while incorporating additional criteria and recent developments in the validation of biogeophysical products. The standardized validation aims to unify validation protocols and datasets to reduce the number of subjective assumptions, thereby improving the transparency, traceability, and comparability of validation results across missions.

The proposed methodology relies on direct validation and product intercomparison approaches:

- The direct validation is performed against ground data set up-scaled according with the CEOS LPV recommendations (Morisette et al., 2006, Fernandes et al., 2014). The confidence in the reference ground-based maps derived from empirical transfer functions depends on performances of the transfer functions that should be quantified with appropriate uncertainty metrics. Two main datasets (DIRECT 2.1 and GBOV) are currently available (Section 4.1.2).
- Intercomparisons with similar remote sensing products (i.e., indirect validation) can determine whether the products behave similarly in space and time on a global scale and allow us to identify differences between products to be investigated in more detail in order to diagnose

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product anomalies and devise algorithm refinements. The LAND VALidation version 2 (LANDVAL V2) network of sites) (Martínez-Sánchez et al., 2024) is used for sampling global conditions in the intercomparison with similar satellite products to avoid using the same BELMANIP2.1 (Weiss et al., 2014) network of sites for training and validation of the successive versions of LAI, FAPAR and FCover products. It provides a global sampling of 2000 locations, representative of global conditions in terms of land cover, continental areas and ecoregion. LANDVAL V2 provides almost identical distribution than the global real distribution of ecosystems (Martínez-Sánchez et al., 2024) improving the global representative of previous LANDVAL V1 (Fuster et al., 2020; Sánchez-Zapero et al., 2023a). To allow comparison between the products, the same temporal (10-days) and spatial (3x3 pixels) supports are used. These analyses are achieved per aggregated land cover class based on the 9 generic classes derived a recent C3S Land Cover V2.1 (https://cds.climate.copernicus.eu/datasets/satellite-land-cover?tab=documentation) for the year 2021 from OLCI and SLSTR data:: Evergreen Broadleaf Forest (EBF), Deciduous Broadleaf Forest (DBF), Needle-Leaf Forest (NLF), Other Forests (OF), Cultivated (CUL), Herbaceous (HER), Shrublands (SHR), Sparse and Bare areas (SBA) and Flooded (FLO).

The same criteria than for the TOC reflectances (Section 5.1) are analysed: completeness, spatial consistency, temporal consistency and Error evaluation (APU and conformity test matching the product requirements). In addition, the stability is evaluated when the length of the data record is typically larger than 10 years.

(1) Product **completeness**.

- Annual global maps of missing values are displayed.
- Distribution of gaps as a function of the season and the length of the gaps are analyzed.

(2) **Spatial consistency**.

- The methodology for visual analysis includes the visualization of zoom over sub-continental areas (i.e., 20º latitude x 30º longitude) at full resolution, and the visualization of animations of global maps at a reduced (1/6 pixels) resolution.
- The spatial consistency is quantitatively assessed by comparing the spatial distribution of a reference validated product with the product biophysical maps under study. Global maps and histograms of residuals, at a reduced (1/6 pixels) resolution, between the product under study and reference products are analyzed in order to identify regions showing spatial inconsistencies for further analysis (e. g. temporal profiles).

(3) **Temporal consistency**.

- Temporal trajectories of the products are qualitatively analyzed over the LANDVAL network of sites plus sites with ground information (i.e., GBOV).
- The cross-correlation metric is the quantitative indicator of the temporal consistency. Distance measures are provided in addition to correlation as they quantify the difference

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between time series, with high sensitivity to amplitude effects (Lhermitte et al., 2011). The most commonly used distance measures in classification or change detection approaches are derived from the Minkowski distance (Jain et al., 1999).

(4) **Error evaluation and conformity testing.**

(5) **Accuracy**, **Precision** and **Uncertainty** (**APU**) are quantified by several metrics (Table 2) reporting the goodness of fit between the products and the corresponding reference dataset. In case of LAI, CEOS LPV recommends RMSD as the overall performance statistic to evaluate the accuracy, due to limitation in the temporal availability of ground datasets (Fernandes et al., 2014). Note that in case of LAI, FAPAR and FCover, two aspects of the precision are also evaluated: **inter-annual** and **intra-annual precision**.

- Scatter-plots and validation metrics (Table 2) versus references are produced. The analysis is complemented with box-plots of Bias and absolute errors per bin.
- Histograms of product values and histograms of residuals per main biome type are evaluated over LANDVAL sites.
- Intra-annual precision (smoothness) corresponds to temporal noise assumed to have no serial correlation within a season. In this case, the anomaly of a variable from the linear estimate based on its neighbours can be characterized as suggested by Weiss et al., (2007): for each triplet of consecutive observations, the absolute value of the difference between the center P(dn+1) and the corresponding linear interpolation between the two extremes P(dn) and P(dn+2) is computed:

$$\boldsymbol{\delta(d_{n+1}) = \left|P(d_{n+1}) - P(d_n) - \frac{P(d_{n+2}) - P(d_n)}{d_{n+2} - d_n}(d_{n+1} - d_n)\right|} \qquad \textbf{Eq. 2}$$

  The distribution of the intra-annual precision is analysed, and the median δ value is used as a quantitative indicator of the inter-annual precision (Fernandes et al. 2014). Hence, the lower the median of δ values, the higher the inter-annual precision.
- Inter-annual precision (i.e., dispersion of LAI values from year to year) can be assessed providing a box-plot of the median absolute deviation of anomalies versus product per bins until the maximum product value (Fernandes et al. 2014). The absolute difference between two consecutive years of the 5th and 95th percentiles of the annual vegetation cycle over the LANDVAL V2 network of sites is then assessed. Note that cultivated sites are not considered due to the non-natural variability of this land cover type induced by agricultural practices (e.g., crop rotation).

In addition**, the compliance of the CLMS uncertainty requirements** is conducted using a simple acceptance (i.e., without considering uncertainties, which are not available) test. Then, the Uncertainty Requirement Compliance (URC) metric is defined as the percentage (%) of match-ups for which |bias| ≤ the uncertainty requirement (optimal or threshold).

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- Conformity testing (simple acceptance): URC (%) for optimal and threshold levels.

(6) Stability

Stability is the property of an instrument to provide similar measurements when the measurand remains constant in time. Analysis of multidecadal long-term stability of satellite data records is a key requirement for the applicability of these products for climate observation. This is of particular importance in data records created from different satellite sensors, as the transition from one sensor to another may create artificial changes in the time series (Mota et al., 2021). While stability assessment has not been addressed in depth for biophysical variables, other EO community has defined stability as the change in bias (apparent error) over a time period (Merchant, 2013).

- Long-term stability is primarily assessed through the analysis of the temporal evolution of the product bias with respect to stable reference datasets, with MODIS V6.1 used as the main reference. The assessment includes the analysis of Hovmöller plots of the bias to identify spatial and temporal patterns, together with the evaluation of temporal trends in the mean bias over a subset of high-quality validation sites with consistent temporal sampling.
- In addition, the temporal evolution of the mean, median, and 5th and 95th percentiles of the retrieved variables is analysed to identify potential shifts in the statistical distribution of the products associated with sensor transitions or other systematic effects.

Table 4 summarizes the quality criteria evaluated, and the corresponding validation metrics and methods that are investigated.

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Table 4: Summary of criteria and validation metrics of LAI, FAPAR and FCover

Criteria	Methods and validation Metrics
Completeness	Gap size distribution (annual maps and temporal variations). Length of gaps.
Spatial Consistency	Visual inspection of global maps and sub-continental areas. Maps and histograms of residuals.
Temporal Consistency	Qualitative inspection of temporal variations. Cross-correlation between time series.
Accuracy	Scatter-plots vs references (N, Bias, MD, R, MAR, p-value). PDFs & histograms of residuals per biome. Box plots of Bias and absolute errors per bin.
Precision	Scatter-plots (STD, MAD). Intra-annual Precision: Histograms of smoothness (δ) and median of 3-point difference Inter-annual Precision: Box-plots per bin and median absolute anomaly of 95th percentile and 5th percentile.
Uncertainty	Scatter-plots (RMSD). Box-plots of absolute bias per bin.
Conformity test	Percentage of pixels matching the CGOS and CGLOPS requirements.
Stability	Hovmöller plots of bias Mean, median and 5th-95th percentiles for different periods over subset of LANDVAL V2 sites

5.3 DMP, GDMP, NPP, AND GPP

The quality assessment of the DMP/GDMP/NPP/GPP 300m V2 product focuses on the following scientific questions:

1. What are the occurrences of quality flags in DMP/GDMP/NPP/GPP 300m V2 in terms of temporal and spatial distribution? Which quality flags are recommended for users and are applied in the subsequent analysis?
2. What are the temporal variation and spatial distribution of the product completeness of DMP/GDMP/NPP/GPP 300m V2? How do the results compare to those of DMP/GDMP 300m V1.0, DMP/GDMP 300m V1.1, DMP/GDMP 1km V2, and MOD17 V6.1 GPP?
3. What are the differences in DMP/GDMP/NPP/GPP 300m V2 between the different consolidation modes RT6, RT2, RT1, and RT0?

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4. How do the DMP/GDMP/NPP/GPP 300m V2 PROBA-V and Sentinel-3/OLCI datasets compare visually to the DMP/GDMP 300m V1.0, DMP/GDMP 300m V1.1, and MOD17 V6.1 GPP datasets (Section 4.2.3).?
5. What is the statistical and spatial consistency between PROBA-V and Sentinel-3/OLCI for DMP/GDMP/NPP/GPP 300m v2?
6. How does DMP/GDMP/NPP/GPP 300m V2 compare with versions DMP/GDMP 300m V1.0 and DMP/GDMP 300m V1.1 spatially and statistically?
7. What is the statistical and spatial consistency with DMP/GDMP 1km V2?
8. What is the statistical and spatial consistency of DMP/GDMP/NPP/GPP 300m V2 in comparison to external MODIS data (MOD17 V6.1 GPP)?
9. What is the statistical and temporal consistency of the DMP/GDMP/NPP/GPP 300m V2 product compared to in-situ FLUXNET GPP data (Section 4.1.3)? Is the uncertainty between DMP/GDMP/NPP/GPP 300m V2 and FLUXNET within the requirement range (Section 2.1)?
10. What is the temporal consistency of DMP/GDMP/NPP/GPP 300m V2 in comparison to MOD17 V6.1 GPP? How do the results of this analysis compare to those of DMP/GDMP 300m V1.0, DMP/GDMP 300m V1.1, and DMP/GDMP 1km V2? What is the stability of the DMP/GDMP/NPP/GPP 300m V2 product?

With the aim of providing answers to the scientific questions, as formulated above, the following protocol is applied on the GPP data (an overview of all procedures is given in Table 5):

(1) Spatial consistency analysis

- *Product completeness*: the percentage of physical GPP values is quantified.
- *Spatial distribution of the GPP estimates*: global maps of metrics expressing the similarity and difference between the yearly CLMS GPP images and reference products are compared. Depending on the source of reference data, the actual yearly data or Short-Term Statistics (STS) are compared.

(2) Global statistical analysis

- *Histograms:* statistical distributions of GPP estimates are computed over biomes for the CLMS DMP data and the STS mean. An aggregated version of the CLMS Global Land Cover at 100m, epoch 2018 (Buchhorn et al., 2019) is used to distinguish between major land cover classes.
- *Scatterplots* between the CLMS GPP product and the reference products are produced at a global scale and per land cover type.
- *Statistical assessment per biome*: metrics among the data CLMS GPP data and the reference products are computed per biome.

(3) Temporal consistency analysis

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- *Temporal variation and realism:* for all sites available, local CLMS GPP temporal profiles are evaluated against in-situ measurements of flux towers.
- *Temporal smoothness*: a smoothness index (Eq. 2) is calculated for each 10-day time step of the archive. A frequency distribution is derived from the time series of smoothness index images.
- *Temporal stability:* the slope of the inter-annual precision of the CLMS GPP product is assessed over all available sites.

(4) Accuracy assessment

- *Direct validation*: the Eddy Covariance in-situ measurements can be considered as ground truth reference data for validating CLMS GPP products. However, it is necessary to bridge the gap in spatial footprint of the two datasets. Each tower is representative for a specific area surrounding the tower. But currently, no detailed information is available to estimate each tower's spatial footprint. In heterogeneous landscapes, this footprint can be quite small (e. g. a small-scale crop parcel) where in homogeneous areas (e. g. a large forest) the measurements of the tower are representative for the surrounding landscape. Also, the wind direction and velocity play a major role. To compare the CLMS GPP products at 300m with the ICOS and Ameriflux datasets, it was decided to extract the single 300m pixel in which each flux tower is located for the CLMS 300m GPP V2 dataset. Scatterplots and validation metrics listed in Table 2 are used to express the accuracy of the CLMS GPP product.

The validation metrics used in the spatial, statistical, and temporal analysis are given in Table 6.

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Table 5: Procedures for the validation of GPP 300m V2.

	Criterium	Method and/or Validation metric	Reference products	Spatial coverage
Q1	/	• Quantification (in %) of flag occurrences over land: temporal and spatial distribution	/	Global subsample
Q2	Product completeness	• Temporal distribution of % of missing values or pixels qualified as 'invalid' over land • Spatial distribution of % of missing values or 'invalid' pixels over land	GPP 300m V1.0 GPP 300m V1.1 GPP 1km V2 MOD17 V6.1 GPP	Global subsample
		• Frequency distribution of the gap length		LANDVAL
Q3	Statistical consistency	• Histogram of overall bias • GM regression analysis	GPP 300m V2 RT0, RT1, RT2	LANDVAL Global subsample on selected tiles
	Temporal consistency	• Temporal pixel profiles		
Q4	Spatial consistency	• Visual comparison	GPP 300m V1.0 GPP 300m V1.1 MOD17 V6.1 GPP	1°x1° tiles (FR) 10°x10° tiles (FR)
Q5	Statistical consistency	• Histogram of overall bias • GM regression analysis, overall and per biome	/	LANDVAL
	Spatial consistency	• Histograms per biome • Spatial distribution of validation metrics		Global subsample
Q6	Statistical consistency	• Histogram of overall bias • GM regression analysis, overall and per biome • Uncertainty requirement analysis	GPP 300m V1.0 GPP 300m V1.1	LANDVAL
	Spatial consistency	• Histograms per biome • Spatial distribution of validation metrics		Global subsample
Q7	Statistical	• Histogram of overall bias	GPP 1km V2	LANDVAL

	consistency	• GM regression analysis • Uncertainty requirement analysis		
	Spatial consistency	• Histograms per biome • Spatial distribution of validation metrics		Global subsample
Q8	Statistical consistency	• Histogram of overall bias • GM regression analysis • Uncertainty requirement analysis	MOD17 V6.1 GPP	LANDVAL
	Spatial consistency	• Histograms per biome • Spatial distribution of validation metrics		Global subsample
Q9	Statistical consistency	• Histogram of overall bias • Uncertainty requirement analysis	FLUXNET	FLUXNET sites
	Temporal consistency	• Temporal pixel profiles		
Q10	Temporal consistency	• Temporal pixel profiles	GPP 300m V1.0 GPP 300m V1.1 GPP 1km V2 MOD17 V6.1 GPP	LANDVAL
		• Frequency distribution of the temporal smoothness		
		• Spatio-temporal evolution of the validation metrics (Hovmöller plots)		Global subsample

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Table 6: Validation metrics used for quality assessment of the GPP and NDVI.

Metric	Description
R^2	The coefficient of determination (R^2) indicates agreement or covariation between two data sets with respect to a linear regression model. It summarizes the total data variation explained by this linear regression model. The result varies between 0 and 1, and higher R^2 values indicate higher covariation between the data sets. In order to detect a systematic difference between the two data sets, the coefficients of the regression line should be used. A disadvantage of R^2 is that it only measures the strength of the relationship between the data, but gives no indication if the data series have similar magnitude (Duveiller et al., 2016a): $$R^2 = \left(\frac{\sigma(X,Y)}{\sigma(X)\cdot\sigma(Y)}\right)^2, \quad \textbf{Eq. 3}$$ with σ(X) and σ(Y) the standard deviation of X and Y and σ(X,Y) the co-variation of X and Y. The R^2 is only provided together with the regression analysis in the global statistical analysis (see below), because it allows a quantitative interpretation of the scatterplots.
GM regression	The geometric mean (GM) regression (GMR) model is used to identify the relationship between two data sets of remote sensing measurements. Because both data sets are subject to noise, it is most appropriate to use an orthogonal (model II) regression like the GM regression. By applying an eigen decomposition to the covariance metrics of X and Y, two eigenvectors are obtained that describe the principal axes of the point cloud (Duveiller et al., 2016b), i.e. the regression line with equation: $$Y = a + b \cdot X \quad \textbf{Eq. 4}$$ The GM regression slope and intercept are added as quantitative information related to the scatterplots. The regression is also used to derive the agreement coefficient and to differentiate between systematic and random differences, as described in the next paragraphs.
Bias	The bias is defined as the actual difference between two datasets: $$bias = X - Y \quad \textbf{Eq. 5}$$
MBE or Accuracy (A)	The Mean Bias Error (MBE) or Accuracy (A) is defined as the actual difference between two datasets and positive and negative differences between observations: $$MBE = \frac{1}{n}\sum_{i=1}^{n}(X_i - Y_i) = \overline{X} - \overline{Y} \quad \textbf{Eq. 6}$$
STD or Precision (P)	The standard deviation (STD) of the bias or Precision (P) is a measure for repeatability: $$STD = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(X_i - Y_i - A)^2} \quad \textbf{Eq. 7}$$
RMSD or Uncertainty (U)	The Root Mean Squared Deviation (RMSD) or Uncertainty (U) measures how far the difference between the two data sets deviates from 0.

	$$RMSD = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(X_i - Y_i)^2}$$ **Eq. 8**
MD	The median bias or median deviation is defined as the median of the difference between two datasets: $$MD = median(X - Y)$$ **Eq. 9**
MAD	The median absolute deviation is the median of the absolute difference between two datasets: $$MAD = median(\lvert X - Y \rvert)$$ **Eq. 10**
Temporal smoothness	The temporal smoothness δ (Weiss et al., 2007) is evaluated by taking three consecutive observations and computing the absolute value of the difference between the centre $P(d_{n+1})$ and the corresponding linear interpolation between the two extremes $P(d_n)$ and $P(d_{n+2})$ as given in Eq.2.
Time series noise	The time series noise can be estimated by averaging δ over the time series (Vermote et al., 2009): $$Noise = \sqrt{\frac{\sum_{n=1}^{N-2}\delta(d_n)^2}{N-2}}$$ **Eq. 11** Relative noise is computed by dividing the noise by the mean of the time series (Claverie et al, 2013): $$Relative\ noise = 100 * \frac{Noise}{\bar{X}}$$ **Eq. 12**
Inter-annual precision	The inter-annual precision is determined as the median absolute deviation of the 5th and 95th percentiles between consecutive years.
Stability	Stability is estimated as the slope of inter-annual precision over time.

5.4 NDVI

The NDVI is not a physical property of the vegetation. Therefore, there are no reference ground measurements. NDVI is a sensor dependent quantity. Still, inter-comparison with NDVI time series derived from other sensors is valuable in order to detect a change in trend between the two NDVI time series.

The quality assessment of the BRDF-corrected NDVI 300m V3 focuses on the following scientific questions:

1. What are the temporal and spatial distributions of the quality flags? Which flags are used in the following analyses and are recommended for users?
2. What are the temporal variation and spatial distribution of the product completeness of NDVI 300m V3? How do the results compare to those of NDVI 300m V1, NDVI 300m V2, NDVI 1km V3, and MCD43A4 V6.1?

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3. How do the NDVI 300m V3 Sentinel-3/OLCI and PROBA-V datasets compare visually to the NDVI 300m V1, NDVI 300m V2, NDVI 1km V3, and MODIS MCD43A4 V6.1 datasets (see Section 4.2.4)?
4. What is the statistical and spatial consistency between PROBA-V and Sentinel-3/OLCI for NDVI 300m V3?
5. How does NDVI 300m V3 compare with the versions NDVI 300m V1 and NDVI 300m V2, spatially and statistically?
6. What is the statistical and spatial consistency with NDVI 1km V3 (including 1km data of SPOT/VGT1, SPOT/VGT2, and PROBA-V)?
7. What is the statistical and spatial consistency of NDVI 300m V3 in comparison with external reference MODIS data (MCD43A4 V6.1)? This allows evaluating the quality of NDVI 300m V3 with respect to the user requirements in terms of acceptable differences with existing satellite-derived products. These requirements are a goal uncertainty of 2.5% and threshold uncertainty of 5%.
8. What is the temporal consistency of NDVI 300m V3 in comparison to MCD43A4 V6.1? How do the results of this analysis compare with those of NDVI 300m V1, NDVI 300m V2, and NDVI 1km V3? What is the stability of the NDVI 300m V3 product?

The overall validation procedures are summarized in Table 7. The procedures are designed to answer the scientific questions (Q1 to Q8) as described above. Depending on the scientific question, emphasis is on statistical, spatial, and/or temporal consistency. Statistical consistency focuses on the distribution of values and the statistical relationship between two datasets. Spatial consistency investigates spatial artefacts, spatial distribution of validation metrics (Table 6), and distribution of values per biome. Temporal consistency focuses on pixel profiles, temporal smoothness, and stability.

The evaluations are done on a systematic subsample of the global images (centre pixel in each window with arbitrary size of 51x51 pixels) in order to have spatial explicit results and to retain global conditions in the data set. This subsample is representative for the global patterns of vegetation and considerably reduces the processing time. In addition, visual comparisons are performed at full resolution for a limited number of 1° x 1° and 10° x 10° tiles. The scatterplots between the different data sets include information on Geometric Mean Regression (GMR) equations and coefficients of determination (R^2). These and other validation metrics that are used for the assessment of statistical and temporal consistency are listed in Table 6. A selection of these validation metrics can be calculated per pixel over time (in order to evaluate spatial consistency) or per time step over grouped pixels (in order to evaluate temporal consistency). The dataset to be validated is X, and the reference datasets are Y.

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Table 7: Procedures for the validation of NDVI 300m V3.

	Criterium	Method and/or Validation metric	Reference products	Spatial coverage
Q1	/	• Quantification (in %) of flag occurrences over land: temporal and spatial distribution	/	Global subsample
Q2	Product completeness	• Temporal distribution of % of missing values or pixels classified as 'invalid' over land • Spatial distribution of % of missing values or pixels classified as 'invalid' over land	NDVI 300m V1/V2 NDVI 1km V3 MCD43A4 V6.1	Global subsample
		• Frequency distribution of the gap length		LANDVAL V2
Q3	Spatial consistency	• Visual comparison	NDVI 300m V1/V2 MCD43A4 V6.1	1°x1° tiles (FR) 10°x10° tiles (FR)
Q4	Statistical consistency	• Histogram of overall bias • GM regression analysis, overall and per biome	/	LANDVAL V2
	Spatial consistency	• Histograms per biome • Spatial distribution of the validation metrics		Global subsample
Q5	Statistical consistency	• Histogram of overall bias • GM regression analysis, overall and per biome • Uncertainty requirement analysis	NDVI 300m V1 NDVI 300m V2	LANDVAL V2
	Spatial consistency	• Histograms per biome • Spatial distribution of the validation metrics		Global subsample
Q6	Statistical consistency	• Histogram of overall bias • GM regression analysis • Uncertainty requirement analysis	NDVI 1km V3	LANDVAL V2
	Spatial consistency	• Histograms per biome • Spatial distribution of the validation metrics		Global subsample

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Q7	Statistical consistency Spatial consistency	• Histogram of overall bias • GM regression analysis • Uncertainty requirement analysis • Histograms per biome • Spatial distribution of the validation metrics	MCD43A4 V6.1	LANDVAL V2 Global subsample
Q8	Temporal consistency	• Temporal pixel profiles • Frequency distribution of the temporal smoothness • Stability requirement analysis • Spatio-temporal evolution of the validation metrics (Hovmöller plots)	MCD43A4 V6.1 NDVI 300m V1/V2 NDVI 1km V3	LANDVAL V2 Global subsample

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5.5 BURNT AREAS

The validation is based on the methods developed in the framework of the ESA CCI Phase II Fire Disturbance project [Fire_cci_D4.1.1_PVR_v2.1]. The reference data used for the validation, covering the years 2020-2022 for the NTC product and 2020 for the NRT product, have been generated by C3S following a supervised semi-automatic algorithm and following the sampling design described in Section 4.1.6. Accuracy measures are based on the error matrix which expresses the amount of agreements and disagreements in terms of area (m^2) between product and reference classifications. Accuracy is measured by ratios between combinations of error matrix cells, as is common in BA validation, by the commission error ratio, the omission error ratio, the Dice coefficient, the absolute bias, and the relative bias. The inferences of accuracy at global scale and for each biome are based on a stratified combined ratio estimator and a total population estimator (Cochran, 1977), depending on the type of accuracy measure, whether it is a ratio or a quantity. Such estimators produce unbiased inferences by taking into account the sampling design, i. e. the size of each stratum, the sample allocation and the amount of area with available data.

In addition to the validation analysis, a quantitative intercomparison analysis is performed taking two global burned areas products, as mentioned in Section 4.2.6. The intercomparison analysis consists of assessing the agreement between each evaluated product and FIRECCIS311, which is used as the reference external product. FIRECCIS311 is the successor to FIRECCIS310, which was identified as the most accurate external product in a validation exercise based on Landsat-derived reference data (Franquesa et al., 2022). Although the intercomparison analyses don't inform about the accuracy of products, they allow a data exploration for entire datasets, not only at relatively small samples as in the case of the validation analyses.

5.6 LAND SURFACE TEMPERATURE

In recent years, the CEOS LPV group has been working towards the systematization of guidelines for LST validation, which are generally also followed here. In line with procedures agreed within the LST community (https://lpvs.gsfc.nasa.gov/LSTE/LSTE_home.html), LST is validated considering: (i) Temperature based validation, i.e., comparisons against in-situ measurements; (ii) Scene-based inter-comparison, i.e., assessment via comparisons with LST products derived from other sensors; (iii) Time series comparisons, mainly to identify possible product degradation over time, or under specific retrieval conditions. It is not foreseen, however, to perform radiance-based validation.

The validation of LST products against in-situ measurements is always a challenging task, since LST is highly variable in space and time. Since most environmental monitoring networks are not specifically designed for satellite LST validation, not all in-situ stations are adequate to assess the quality of the CLMS LST product. Stations within each network are equipped with a similar set of

instruments. However, not all of them provide the required measurements to assess LST, which include both up-welling and downwelling longwave radiation fluxes. KIT has deployed a few stations over highly homogeneous sites equipped with appropriate and well-calibrated sensors: Gobabeb and Farm Heimat, Namibia; Evora, Portugal and Lake Constance in the Germany/Switzerland border. However, most of these stations do not report their data in near real time, making them difficult to use in yearly evaluations. The representativeness of LST at pixel scale by station measurements is a critical issue. In-situ measurements that do not appropriately represent pixel LST should be avoided, although often compromises are needed due to the scarcity of local observations. In order to assess the suitability of a given station for satellite LST validation, its surroundings need to be carefully examined, namely using high resolution information on orography, land cover and LST retrievals performed with high resolution sensors such as Landsat. The Google Earth Engine framework (Gorelick et al., 2017) and the package to derive LST from Landsat retrievals, developed by Ermida et al. (2020) have proven useful in handling these tasks. Validation of new LST products/product versions should rely on as many stations as possible, despite taking into account the limitations described above. The regular assessment of LST operational products/product versions aims to provide information on the stability of the products' performance and therefore should focus on a relatively small number of reference stations covering a representative range of surface and climate conditions – ideally these should be identified by GBOV.

Satellite/in-situ matchups are performed using the central pixel of the original 5x5 retrieval window closest to each GBOV station. Only observations with an absolute acquisition time difference of 3 minutes are considered for stations with 1–5 minute sampling intervals (Trigo et al., 2008), while for stations with 30-minute sampling intervals, the nearest available satellite observation is selected. The resulting matchups are evaluated independently at each station using standard statistical metrics to quantify the agreement between satellite-derived and in-situ estimates.

The direct validation exercise includes a comparison between CLMS LST V3.0 and GBOV observations, considering: (i) scatterplots of satellite-derived LST estimates versus in-situ observations and; (ii) a set of robust statistical metrics recommended by the CEOS LPV Working Group for LST Validation (Guillevic et al., 2018), including bias (reporting on accuracy), absolute residuals (reporting on precision), and root mean square differences (reporting on magnitude and error distribution). The evaluation of the LST product against the contract specifications (adapted from GCOS-245) shall be performed via conformity tests that measure the percentage of matchups between the LST and each in-situ observations that conform (if the absolute error is lower or equal than the maximum permissible error) or non-conform (if the absolute error is higher than the maximum permissible error). The VR shall cover ideally one year or more and a wide range of surface and atmospheric conditions, allowing a comprehensive characterization of LST uncertainties. Additional consistency analyses are performed to complement the in-situ validation by assessing the temporal and spatial consistency of the CLMS LST V3.0 product. These analyses use independent

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reference datasets and complementary assessment approaches to characterize the stability of V3.0 and identify differences associated with the retrieval, input datasets and processing configuration.

Temporal stability is assessed through comparisons between CLMS LST V3.0 and the ERA5 skin temperature reanalysis dataset (Section 4.3.2). To ensure consistent comparisons between datasets, the CLMS LST V3.0 is spatially aggregated to match ERA5's regular 0.25º latitude–longitude grid. For each grid cell, hourly LST values are used to derive the monthly mean diurnal cycle, from which the minimum, mean, and maximum LST values are extracted, provided that at least 30 % of the hourly observations are valid. These quantities are then averaged to obtain monthly mean values and, subsequently, monthly anomaly differences. The latter, are calculated by removing the mean seasonal cycle from both time-series. The analysis is performed at global and regional scales to characterize the temporal behaviour of CLMS LST product across the different geostationary domains (GOES, MSG-IODC, and Himawari).

The consistency of the CLMS LST product is further evaluated using the Sentinel-3A SLSTR LST product from the ESA CCI LST dataset (https://climate.esa.int/en/projects/land-surface-temperature/) (Section 4.2.5). Sentinel-3A SLSTR Level-3 supercollated global datasets (version 4.00; Ghent et al., 2025) are used as an independent satellite-based reference and aggregated to the spatial grid of the CLMS LST product. The comparisons are performed using temporally and spatially collocated daytime and nighttime observations, retaining only pixels with valid observations in both datasets. Monthly anomaly differences between CLMS LST V3.0 and Sentinel-3A SLSTR LST are then computed to evaluate their consistency over the common study period. It should be kept in mind that such comparisons are conditioned by 1) differences in the sensor spatial resolutions, 2) by the differences viewing-illumination geometry, and 3) by differences in the algorithms and in their calibrations.

Spatial consistency is assessed through qualitative analyses of seasonal mean LST difference maps against other reference datasets (e.g. previous CLMS LST product versions), when applicable. These comparisons allow the characterization of regional spatial variability and the identification of patterns related to surface conditions, observation geometry, and retrieval configurations.

Additionally, qualitative case studies of severe heatwave events can be used to illustrate the capability of the LST10 Synthesis and TCI products to represent the spatial distribution and temporal evolution of thermal stress, highlighting the added value of the 10-daily products for operational monitoring applications.

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5.7 SURFACE SOIL MOISTURE AND SOIL WATER INDEX

The quality assessment of the Surface Soil Moisture (SSM) and of the Soil Water Index (SWI) is in general carried out in the same way, as both represent the same variable soil moisture (SM). In preparation for the quality assessment, the daily CLMS products are converted to daily time series and masked for invalidating conditions using the quality flagging provided by the ISMN in-situ reference for frozen soil, snow cover, and other invalidating conditions (Section 4.1.5). The selected reference time series are temporally matched to the SM observation timestamps using temporal nearest neighbour matching. This is necessary since the CLMS SM data are delivered as daily dataset whereas the reference datasets have commonly multiple observations per day.

Since the CLMS SM data is provided in percentage of relative saturation, while the reference products are provided in m^3/m^3, data scaling is necessary prior to the evaluation of difference metrics such as RMSD, and scaling into the in-situ data space is commonly done in the literature (Brocca et al., 2011). In order to report here the targeted metrics in m^3/m^3 and to ensure consistent results across reference products, "mean-standard-deviation scaling" (also known as z-normalisation) is used to scale the CLMS SM data to the reference products (Lima and Souza, 2023). With this method, the CLMS SM data is scaled to have the same mean value and standard deviation as the reference dataset. This approach was chosen because it is widely adapted by the community and because the extent of the scaled values is less affected by the presence of outliers as would be the case with using "minimum-maximum scaling".

The coarse-scale, global SWI products (that is solely based on Metop ASCAT observations) are compared to in-situ data (e. g. from the International Soil Moisture Network, ISMN, described in Section 4.1.5) and suitable reanalysis model output (e.g. from GLDAS-Noah V2.1 described in Section 4.3.1) in temporal analyses. The soil moisture values of the SWI are compared to different soil moisture layers of the reference datasets, with basic metrics as Pearson's correlation coefficient (rho) with a p-value < 0.05, and Root Mean Square Difference (RMSD).

The higher resolution products (SSM1km and SWI1km) undergo the same temporal analysis as the coarse scale product, using both in-situ and modelled reference data. Comprehensive model data at a similar spatial resolution is generally not available for the whole extent of the 1km products. However, soil moisture dynamics are largely driven by atmospheric forcing that acts at much coarser scales. Coarse scale model data therefore provides a robust reference for the temporal signal of the 1km products, in particular for statistics aggregated over many locations. The comparison against in-situ data in turn contributes actual ground measurements, whose limited spatial representativeness needs to be taken into account (see below). Both reference types thus offer complementary views on the product quality, and are analysed together to obtain a more complete picture than either could provide alone.

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Interpretation of Metrics

The validation metrics need to be interpreted jointly, as each metric alone is blind to certain error characteristics. In particular, difference metrics such as the RMSD and its unbiased variant (ubRMSD) scale with the variance of the compared time series. For example, a malfunctioning in-situ sensor that reports near-constant values yields a deceptively small ubRMSD. Likewise, locations with weak soil moisture dynamics generally yield smaller ubRMSD values than locations with pronounced dynamics. Still, the ubRMSD is a valuable metric, as it reports errors in the physical unit of the reference data (m^3/m^3), which eases interpretation for many users. However, ubRMSD values must not be compared against fixed accuracy thresholds without regarding the variance and quality of the reference data.

To ensure robust results, all reported metrics are subject to the following requirements: in-situ reference data is screened using the ISMN quality flags (Section 4.1.5), correlation values are only reported when statistically significant (`p-value < 0.05`), and time series enter the analysis only with a sufficient number of valid, temporally matched observations.

Spatial Representativeness

For the quality assessment of the CLMS SM products, actual ground measurements from in-situ stations are of major importance. When comparing remotely sensed and in-situ data, one must consider spatial representativeness. This means how well an in-situ observation, which is essentially a point measurement, represents the data gathered by the much larger satellite footprint. This is in general different for each station and depends on topography, soil types and microclimates. De Lannoy et al. (2007) discussed relationship between point measurements and the average behaviour of field-scale soil moisture and concluded that the choice of individual sites has considerable impact on validation campaigns, suggesting field-level analysis of the local spatial SM signal structure and its drivers, followed by upscaling procedures. For the quality assessment activities within CLMS, all datasets are kept with their native product samplings as an in-depth analysis of in-situ sites is out of scope in this endeavour. Consequently, results from this validation need always be understood in reflection of the large difference of scales.

While for the coarse scale data (i. e. with a 0.1° sampling) temporal soil moisture dynamics dominate area-wide and are not controlled by local land cover heterogeneity, for the 1km-scale data this so-called representativeness error is assumed not negligible and thus quality assessment results need to be reflected accordingly. Crow et al. (2012) stated that during validation activities based on comparisons between point-like ground observations and extensive satellite retrievals, this sampling error can be misattributed to retrieval uncertainty and spuriously degrade the perceived accuracy of satellite soil moisture products. They found that standard deviations of SM in sample sets spanning $0.8km^2$ and $1.6km^2$ are of the same or larger magnitude as the CEOS' accuracy criterion for SM of

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0.04 m³/m³. Gruber et al. (2013) carried out an in-depth analysis of coarse scale representativeness of ISMN networks and they found absolute error estimates ranging from 0.02 to 0.06 m³/m³. However, one can expect general lower errors when analysing SM anomalies and relative dynamics. Consequently, differences between point-scale in-situ observations and the kilometric scale products must not be attributed to product retrieval errors alone. Alternatively, a validation against model SM data of the same scale as the satellite product could be done. However, for a continentally or globally comprehensive quality assessment of remotely sensed SM at kilometric scale, there is a lack of comparable extensive datasets with similar spatial resolution and extent. Fortunately, regional model output at kilometric scale describing SM, even with limited extent, can provide an accurate assessment of the temporal and spatial SSM signal quality. The reader may here refer to the studies of Bauer-Marschallinger et al. (2018) and Bauer-Marschallinger et al. (2019), where the authors examined a SWI1km dataset against an extensive 1km soil moisture model dataset covering the Umbria region in Italy.

Potential Additions

The comprehensive quality assessment of remotely sensed soil moisture at kilometric scale is hampered by the lack of comparable extensive datasets with similar spatial resolution. Some in-situ soil moisture networks comprise data from stations that cover multiple adjacent kilometric scale pixels. These could be exploited for the quality assessment of fine scale spatial differences. Moreover, model output at kilometric scale describing soil moisture, even with limited extent, may provide an accurate assessment of the temporal and spatial SSM signal quality. Thus, both assessments of the temporal and the spatial quality could be complemented by comparisons to suitable high-resolution reference data.

As certain quality differences between locations are caused by the dominant land cover in the pixel, a land cover-based quality assessment could be performed. Therefore, the same spatial and temporal analysis as described above could be conducted, but on subsets of the datasets, e.g., using only cropland, grassland, or forest pixels. Thus, an additional component can be added to the quality assessment that will both increase the data usability for the users as well as support the identification of future improvements of the product algorithms.

Moreover, advanced validation metrics based on triple collocation analysis, such as the signal-to-noise ratio (SNR, Gruber et al., 2016 and Gruber et al., 2020), could complement the traditional metrics in future quality assessment cycles. These metrics estimate the random error of each dataset without assuming an error-free reference. However, they are established for coarse resolution soil moisture products and rest on assumptions on error independence and data quality that are not readily fulfilled at kilometric scale. Their application to the 1km products therefore requires a prior, validated adaptation of the methodology, in concert with suitable in-situ and model reference data. Premature application may yield misleading error estimates.

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5.8 LAND SURFACE PHENOLOGY

The validation procedures are mainly based on the CEOS LPV validation protocol for biophysical variables (Fernandes et al., 2014) and on the state-of-the-art for validation of satellite-based phenology products (e.g., Rodriguez-Galiano et al., 2015; Melaas et al., 2016; Peng et al., 2017; Zhang et al., 2018b; Bolton et al., 2020, Moon et al., 2021). The quality assessment relies on direct validation and product intercomparison approaches.

- The **direct validation,** comparing satellite products to ground measurements of the corresponding biophysical variables. The direct validation is usually limited by a small number of reference sites. PEP725, TEMPO, NEON, and PhenoCam V3 datasets (Section 4.1.7) are used to validate the start and end of the growing season. Note that PEP725, TEMPO and NEON provide phenological observations whereas in PhenoCam phenophases are indirectly derived from cameras.
- The **indirect validation**, including intercomparison with similar or using other suitable reference datasets. The indirect validation allows extending the evaluation over large areas or time periods but only relative discrepancies between existing datasets can be assessed. Reference global products providing phenology metrics (MOD12Q2 V6.1 and VNP22Q2 V2, Section 4.2.7) and productivity parameters (CLMS DMP/GDMP and MOD17, Sections 2.3 and 4.2.3) are used, as well as the European CLMS HR-VPP product (Section 4.2.7). Two different samplings are used depending on the reference product:
    - For global products, the LAND VALidation version 2 (LANDVAL V2) network of sites is used (Martínez-Sánchez et al., 2024), which provides a global sampling representative of global conditions in term of land cover, continental area and ecoregion. LANDVAL V2 counts with a total of 2000 sites and is based on Copernicus Global Land Cover based on PROBA-V (Buchhorn et al., 2019) and World Wild LiFe Terrestrial Ecoregion of the World classification (Olson et al., 2001). The following quality criteria to select the coordinates along the globe are: i) same land cover and ecoregion (>90% homogeneity) over an area of 3 km x 3 km; ii) distance between samples at least of 2 degrees on the hypotenuse; and iii) slope within the 3 km x 3 km area lower than 100 m.
    - For European products, the EVAL V2 network composed by 3800 evaluation sites representative of the European region is used. This sampling was previously used and defined in the validation report of HR-VPP products [CLMS_HRVPP_VR_ST-VPP]. Sites were defined to be homogeneous over areas of, at least, 700 m^2, and are are well distributed across the European region.

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To allow comparison between the products, the same spatial (1 km^2) support is used. These analyses are achieved per aggregated land cover class based on the 8 generic classes derived from the Global Land Cover 100m (GLC-100m) classification (Buchhorn et al., 2020): Evergreen Broadleaf Forest (EBF), Deciduous Broadleaf Forest (DBF), Needle-Leaf Forest (NLF), Other Forests (OF), Cultivated (CUL), Herbaceous (HER), Shrublands (SHR) and Sparse and Bare areas (SBA).

The following criteria are analysed: product completeness, spatial consistency and error evaluation, which involves APU. In addition, the conformity test matching the CGLOPS product requirements is also performed.

(1) Product completeness.

- Yearly global maps and percentages of valid and missing data are provided.

(2) Spatial consistency.

- The methodology for visual analysis includes the visualization of the global maps of the product content at reduced resolution (i.e., 1/48), and the zooms over sub-continental areas (i.e., 10º latitude x 15º longitude) and Areas of Interest (AOIs) of typically between 5 km to 5 km to 20 km x 20 km.
- The spatial consistency with reference products includes:
  - Global maps and histograms of differences with global reference satellite products. Furthermore, global distribution of pixels within the predefined uncertainty levels.
  - Comparison with HR-VPP over the AOIs.
- Additionally, the phenological trend as a function of the latitude is analysed. A latitudinal transect is defined as the geodesic line from northern to southern latitudes from Scandinavia to South Africa. The spatial variability is computed using the STD and MAD metrics (i.e., indicative of precision) over selected homogeneous AOIs.

(3) Error evaluation:

- APU metrics are quantified by several metrics (Table 2) reporting the goodness of fit between the products and the corresponding reference dataset. Scatter-plots and validation metrics versus references are produced. The analysis is complemented per biome type.
- Finally, the conformity test is provided, which corresponds to the computation of the percentage of pixels within the product user requirements.

Table 8 summarizes the quality criteria evaluated, and the corresponding validation metrics and methods that are investigated.

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Table 8: Summary of criteria and validation metrics of LSP.

Criteria	Methods and validation Metrics
Completeness	Global maps of valid and missing data.
Spatial Consistency	Visual inspection of global maps, sub-continental areas and local zooms (AOIs) of all product layers. Maps and histograms of differences. Phenological trend as a function of the latitude. Spatial variability (precision: STD, MAD) over AOIs.
Accuracy	Scatter-plots vs references (N, Bias, ME, R, MAR, p-value). Analysis per biome type.
Precision	Scatter-plots (STD, MAD).
Uncertainty	Scatter-plots (RMSD).
Conformity test	Percentage of pixels matching the CGLOPS requirements.

5.9 Evapotranspiration and Heat Fluxes

The quality assessment of the ETA v1.0 and HF v1.0 products consists of a direct and an indirect validation. The direct validation refers to the comparison with actual evapotranspiration (ET), surface latent heat (LE) and sensible (H) values derived from high frequency in-situ measurements of surface energy fluxes by sensors installed at eddy covariance towers (Section 4.1.3). A number of statistical metrics were derived from such comparison which can then be grouped by climate region, plant functional type or other relevant grouping criteria. The metrics included: the Pearson correlation coefficient, bias, relative bias, root mean squared error and relative root mean squared error. The latter informed about the uncertainty of the product. The actual derivation of statistics was preceded by a step in which the original eddy covariance datasets was aggregated to match the dekadal time step at which the CLMS ETA product is delivered. For the HF validation, the nearest by eddy covariance timeslot of half-hourly measurements was chosen.

The indirect validation relies on the comparison of other products (Section 4.2.8) containing the same variables as in the CLMS product. Note that the use of external products in this validation exercise does not imply that those products are considered the truth and/or an absolute reference. The value of the indirect validation is the possibility of verifying similarities and/or detecting differences in major spatial and temporal patterns in the CLMS products when compared to similar products generated through different methods and exploiting different forcing datasets. The comparison

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examines temporal and spatial patterns across different spatial windows and/or latitudinal transects. Unlike the direct validation, this analysis was only performed for the year 2021.

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